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Differences in size and number of embryonic type-II neuroblast lineages correlate with divergent timing of central complex development between beetle and fly

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eLife Assessment

The study is a **valuable** contribution to the question of evolutionary shifts in neuronal proliferation patterns and the timing of developmental progressions. The authors present **convincing** data which confirm the presence of type-II NB lineages in beetle with the same molecular characteristics as the *Drosophila* counterparts but differing in lineage size and number. The data lay the foundation for future analysis of the role and molecular characteristics of individual lineages and of whether differences in the identity, proliferation pattern and timing of developmental progression can be linked to differences in the development of functionality of the central complex.

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Abstract

Despite its conserved basic structure, the morphology of the insect brain and the timing of its development underwent evolutionary adaptations. However, little is known about the developmental processes that create this diversity. The central complex is a brain centre required for multimodal information processing and an excellent model to understand neural development and divergence. It is produced in large parts by type-II neuroblasts, which produce intermediate progenitors, another type of cycling precursor, to increase their neural progeny. These neural stem cells are believed to be conserved among insects, but little is known on their molecular characteristics in insects other than flies. *Tribolium castaneum* has emerged as a valuable new insect model for brain development and evolution. However, type-II neuroblast lineages and their role in central complex development have so far not been studied in this beetle.

Using CRISPR-Cas9 we created a fluorescent enhancer trap marking expression of *Tribolium fez/earmuff*, a key marker for type-II neuroblast derived intermediate progenitors. Using

combinatorial labelling of further markers including *Tc-pointed*, *Tc-deadpan*, *Tc-asense* and *Tc-prospero* we characterized the type-II neuroblast lineages present in the *Tribolium* embryo and their sub-cell-types. Intriguingly, we found 9 type-II neuroblast lineages per hemisphere in the *Tribolium* embryo while *Drosophila* produces only 8 per brain hemisphere. In addition, these lineages are significantly larger at the embryonic stage of *Tribolium* than they are in *Drosophila* and contain more intermediate progenitors. Finally, we mapped these lineages to the domains of early expressed head patterning genes. Notably, *Tc-otd* is absent from all type-II neuroblasts and intermediate progenitors, whereas *Tc-six3* marks an anterior subset of the type-II-lineages. The placodal marker *Tc-six4* specifically marks the territory where anterior medial type-II neuroblasts differentiate.

In conclusion, we identified a conserved pattern of gene expression in holometabolan central complex forming type-II neuroblast lineages, and conserved head patterning genes emerged as new candidates for conferring spatial identity to individual lineages. The higher number and greater lineage size of the embryonic type-II neuroblasts in the beetle correlate with a previously described embryonic phase of central complex formation which is not found in the fly. These findings stipulate further research on the causal link between timing of stem cell activity and temporal and structural differences in central complex development.

Introduction

Neural development of insects allows to study molecular and cellular principles in easy to manipulate invertebrate models. The fruit fly *Drosophila melanogaster* (*Drosophila* hereafter) has mostly been used for this purpose, making it one of the best understood models for neurogenesis [1,2]. However, it has remained a puzzle how the huge ecological diversity of insects and the divergent neural anatomies that are adapted to different niches [3,4] evolved. Furthermore, *Drosophila* may in many instances not be a good representation of insect development and some processes are derived in the fly lineage [5–7]. For these reasons the beetle *Tribolium castaneum* (*Tribolium* hereafter) has been introduced as an additional model for insect neural development [8–11]. *Tribolium* is a grain storage pest and all stages live in flour, hence there is no major transition of habitat or lifestyle between the larval and adult stage as in flies. However, when compared to *Drosophila* the larvae are more mobile through the use of walking legs which allows them to navigate within or between food sources [3,6,10]. In this beetle many molecular genetic manipulation and labelling methods have been established [12–16]. Specifically, labelling of neural cell types in *Tribolium* has been facilitated by the advent of CRISPR-Cas9 [12,15,17]. Another factor that makes *Tribolium* an informative model for the understanding of insect brain development is the more conserved development of the head neuroectoderm facilitating the identification of specific neurogenic domains and allowing cross-species comparisons [11,18,19].

The insect central complex is an anterior, midline spanning neuropile and constitutes an important brain centre for the processing of sensory input, coordination of movement and navigation [20,21]. The highly conserved basic structure of the central complex, along with diverse specifications in response to ecological requirements among different insect species, make this brain structure a very interesting model for evolutionary developmental studies [3,10,20]. Between the two insect models *Drosophila* and *Tribolium* a temporal shift in the emergence of central complex neuropile was observed with *Tribolium* developing a functional larval central complex during embryogenesis whereas it develops only at the onset of adult life in *Drosophila* [3,10]. It is however not known how these temporal differences are established during development on a cellular and molecular level.

The entire insect central nervous system including the brain is produced by neural stem cells, the neuroblasts (NBs) [22]. While there are evolutionary modifications in the relative position of trunk NBs and their gene expression profiles between fly and beetle, the core determinants specifying their role as neural progenitors are conserved [8]. NBs undergo repeated divisions producing rows of ganglion mother cells (GMCs) which divide one more time to produce neurons and/or glia (see Fig. 1, top panel). This leads to neural cell lineages that include the neuroblast itself and its progeny [23,24]. Several NBs of the anterior-most part of the neuroectoderm contribute to the central complex, which has been characterized as the most complex brain structure (apart from the optic lobe) [25,26]. Intriguingly, in *Drosophila* and in the grasshopper *Schistocerca gregaria* a neuroblast subtype, the type-II neuroblasts (type-II NBs), were found to prominently contribute to the formation of the central complex [27,28]. These specific neuroblasts generate more offspring by producing another class of neural precursors, the intermediate progenitors (INPs), which also divide in a stem cell like fashion [29,30] (Fig. 1, lower panel). Type-II NBs and INPs have attracted a lot of attention because intermediate, cycling progenitors have also been described in vertebrates, the radial glia cells, which produce a variety of cell types [31,32]. In addition, in *Drosophila* brain tumours have been induced from type-II NB lineages [33], opening up the possibility of modelling tumorigenesis in an invertebrate brain, thus making these lineages one of the most intriguing stem cell model in invertebrates [34,35].

In the developing insect brain the number of neurons that is produced by each type-II NB is significantly larger than the offspring of a type-I NB. Medial *Drosophila* type-II NBs produce an average of 450 neurons whereas type-I NBs of the brain only generate 100-200 neurons each [37–39]. This increased number of neural cells that are produced by individual type-II lineages along with the fact that neuroblast lineages can produce different types of neurons, leads to the generation of extensive neural complexity within the anterior insect brain [27,30].

Based on descriptions of large proliferative lineages in the grasshopper *Schistocerca gregaria*, a hemimetabolous insect distantly related to flies, it is widely believed that type-II NBs and INPs are conserved within insects [25,40]. However, a molecular characterisation of such lineages that on the one hand will allow to test whether the cell type markers characterized in *Drosophila* are more widely conserved, and on the other hand will facilitate an identification of sub-cell types, has not been performed in an insect other than *Drosophila*.

Whereas it is widely believed that *Tribolium* does use type-II NB in central complex formation [3,17,41] their presence has so far not been unequivocally shown and a thorough comparison of type-II NB lineages and their sub-cell-types between the fly and the beetle model has been lacking.

The characterization of type-II NBs in *Drosophila* has in large parts been based on a specific reporter line in which eGFP expression is driven by an *earmuff* (*fez/erm*) enhancer element and marks INP-lineages [30,42,43]. *Tribolium earmuff* has been first described as *Tc-fez*, referring to the vertebrate *earmuff* ortholog *fez* [19]. We will use *Tc-fez/erm* in the following. A defined sequence of gene expression has been described in *Drosophila* type-II NBs, INPs and GMCs. The ETS-transcription factor *pointed* (*pnt*) marks type-II NBs [44,45], which do not express the type-I NB marker *asense* (*ase*) but the neural gene *deadpan* (*dpn*) (Fig. 1). Intermediate progenitors express *fez/erm* and *ase* and at a more mature stage also *dpn*, whereas *prospero* (*pros*) marks mature INPs and GMCs [30,46]. *Six4* is another marker for *Drosophila* type-II NBs and INPs [47].

In the present work we characterize type-II neuroblasts in *Tribolium* with respect to their number, their location and the conservation of the molecular markers known from *Drosophila*. We also wanted to test, in how far the earlier emergence of the central complex in beetles would be reflected by a change of division activity of type-II NBs or intermediate progenitors. To that end,

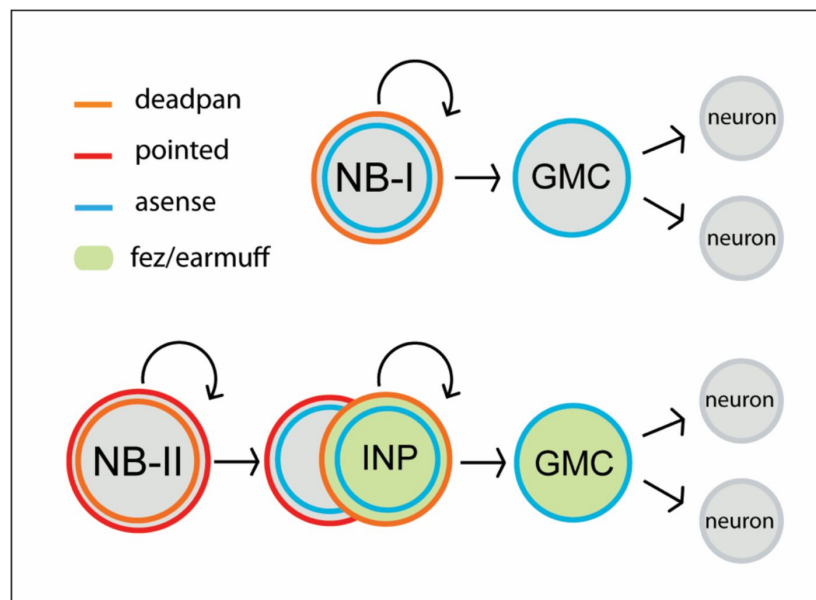


Figure 1

Neuroblasts of the *Drosophila* nervous system.

Type-I neuroblasts (type-I NBs) (top panel) and type-II neuroblasts (type-II NBs) (lower panel). Type-I NBs undergo stem-cell-like divisions with each round producing a ganglion mother cell (GMC) that divides one more time to produce two neurons or glia cells. In *Drosophila*, all type-I NBs express *dpn* and *ase* whereas GMCs express only *ase*. Type-II NBs express *dpn* and *pnt* but not *ase*. They also undergo repeated divisions, with each round producing an intermediate progenitor cell (INP) which also is a proliferating progenitor and expresses *ase*. INPs undergo a maturation process during which *pnt* is still expressed initially, but not *dpn*. Mature INPs express *dpn*, *ase* and *fez/erm* whereas GMCs express *ase* and *fez/erm*. Type-II NB lineages are only found in the anterior brain, they produce central complex neurons and glia [30,36].

we used CRISPR-Cas9 induced genome editing for creating a *Tc-fez/erm* enhancer trap line to characterize embryonic *Tc-fez/erm*-expressing cells including INPs and GMCs of type-II NB lineages and combined this line with other labelling methods including multicolour *hybridisation chain reaction* (HCR) [48,49]. We found that the *Tribolium* embryo produces 9 *Tc-pnt*-expressing type-II NBs compared to only 8 type-II NB lineages in the *Drosophila* embryo [50]. We show that these lineages produce central complex cells and confirm a largely conserved molecular code that differentiates INPs of distinct maturation stages and GMCs. Intriguingly, we found that lineage sizes in *Tribolium* embryos are considerably larger and include more mature INPs than in the *Drosophila* embryo [50]. We also show that the placodal marker gene *Tc-six4* characterizes the embryonic tissue that gives rise to the larger anterior group of type-II NBs, whereas *Tc-six3* is marking only an anterior subset of these lineages. Interestingly there is a part of the central complex precursors expressing neither *Tc-six3* nor *Tc-otd*, which is absent from all type-II NB lineages.

Results

A CRISPR-Cas9-NHEJ generated enhancer trap lines marks *Tc-fez/erm*-expressing cells at the embryonic and larval stage

We created an enhancer trap line driving eGFP that reflects *Tc-fez/erm* gene expression in the embryo. We named the line *fez magic mushrooms* (*fez-mm-eGFP*) because it marks the mushroom bodies of the larval brain (Fig. 2 A-C). The reporter construct was inserted 160 bp upstream of the *fez/erm*-transcription start site using the *non-homologous end joining* (NHEJ) repair mechanism (see supplementary Fig. S1 for scheme). Enhancer traps sometimes only reflect parts of a gene expression pattern, therefore we carefully evaluated eGFP co-expression with *Tc-fez/erm*-RNA. We found that at the embryonic stage *fez-mm-eGFP* co-localized with most *Tc-fez/erm*-expressing cells in an area that will give rise to the protocerebral structures mushroom bodies, larval eyes and central complex [11,19] (Fig. 2, panel D). The *fez-mm-eGFP* line marked several progenitor cell types including cells that we characterize as type-II NB-derived intermediate progenitors (see below). Based on these analyses, we use the *fez-mm-eGFP* expression as a reporter for *Tc-fez/erm* expression.

Tc-pointed marks a population of nine type-II NBs that are associated with *fez-mm-eGFP* marked INP-lineages

To identify type-II NBs, we searched for cell groups that express the markers known from flies. The transcription factor *pointed* (*pnt*) marks type-II NBs in *Drosophila* and is required for their correct differentiation [44,45]. In both *Tribolium* and *Drosophila*, *pnt* is also expressed in other areas (this work/[51,52]). Therefore, we looked for *Tc-pnt* expression in large cells with large nuclei (typical NB morphology) that were closely associated with *Tc-fez/erm*-expressing cells (marking putative INPs). We found a total of 9 such clusters instead of the 8 expected from flies (Fig. 3 A-C). We also found that the large *Tc-pnt* expressing cells as well as *fez-mm-eGFP* expressing cells are mitotically active (Fig. 3 C-I). Further molecular and cell size analysis corroborated our interpretation that these cells constitute type-II NBs, intermediate progenitors (INPs) at different maturation stages and ganglion mother cells (GMCs) (see details below).

Looking at a series of embryonic stages we determined the emergence and further development of *Tc-pnt* and *fez-mm-eGFP* expressing lineages as well as their arrangement with respect to the embryonic head and to one another. We first detected conspicuous adjacent expression of *Tc-pnt* and *fez-mm-eGFP* at stage NS7 (not shown). At stage NS11 seven *Tc-pnt*-clusters are present in a horse-shoe-shaped formation surrounding *fez-mm-eGFP* positive cells (white arrows in Fig. 3 A, star marks one cluster out of focus). At stage NS13 these groups are found in a position at the

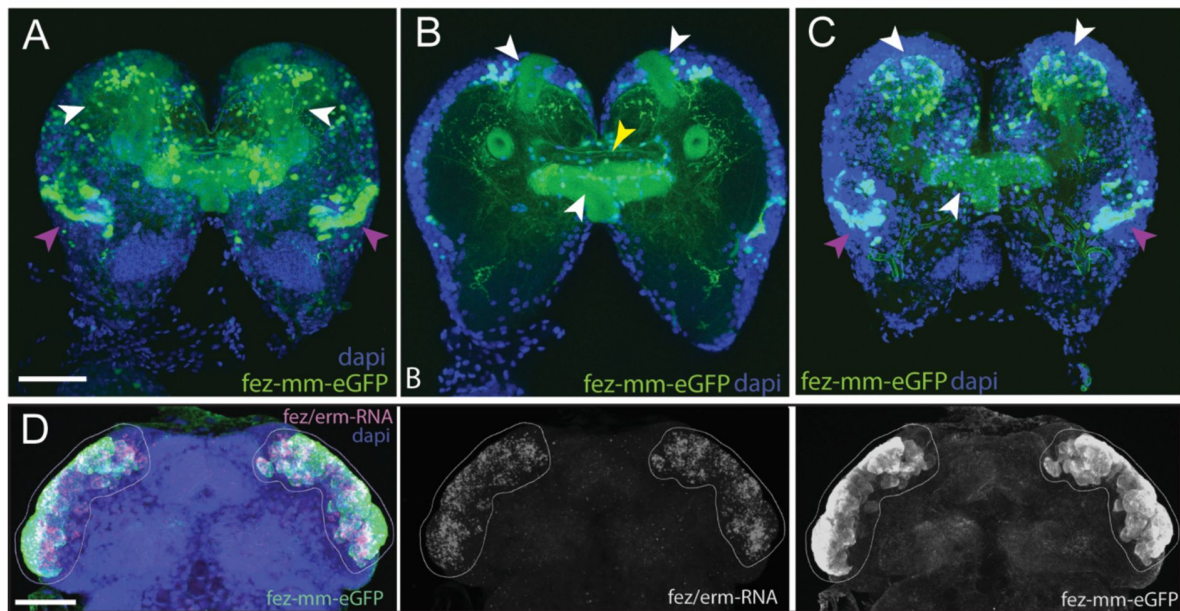


Figure 2

Magic mushrooms (*fez-mm-eGFP*) reporter line.

A-C) eGFP-expression in a larval instar brain, D) EGFP antibody labelling in combination with *Tc-fez/erm* RNA *in situ* in the embryonic head, A-D) Dapi labelling of nuclei. Scalebars in A and D are 50 μm. **A)** Full projection of whole brain scan (4th instar larvae): *fez-mm-eGFP* expression marks many cells of the mushroom bodies (white arrowheads) and of the larval optic lobe (purple arrowheads). Note that we use the terms dorsal and ventral with respect to the neuraxis of the larva. See [10] for orientation. **B)** Central part of the whole brain scan (projection of substack, same brain as in A): *mm-eGFP* marks the peduncles of the mushroom bodies (white arrowheads) and some central complex ensheathing cells (probably glia cells; yellow arrowhead). **C)** Projection of the dorsal part of same brain as in A shows calyces and peduncles of mushroom bodies (white arrowheads), and optic neuropile (purple arrowheads). **Panel D)** Extensive co-localization of *fez-mm-eGFP* with *Tc-fez/erm*-RNA in white encircled area which gives rise to protocerebral structures. Projection of all planes where expression was detected, embryonic stage NS11.

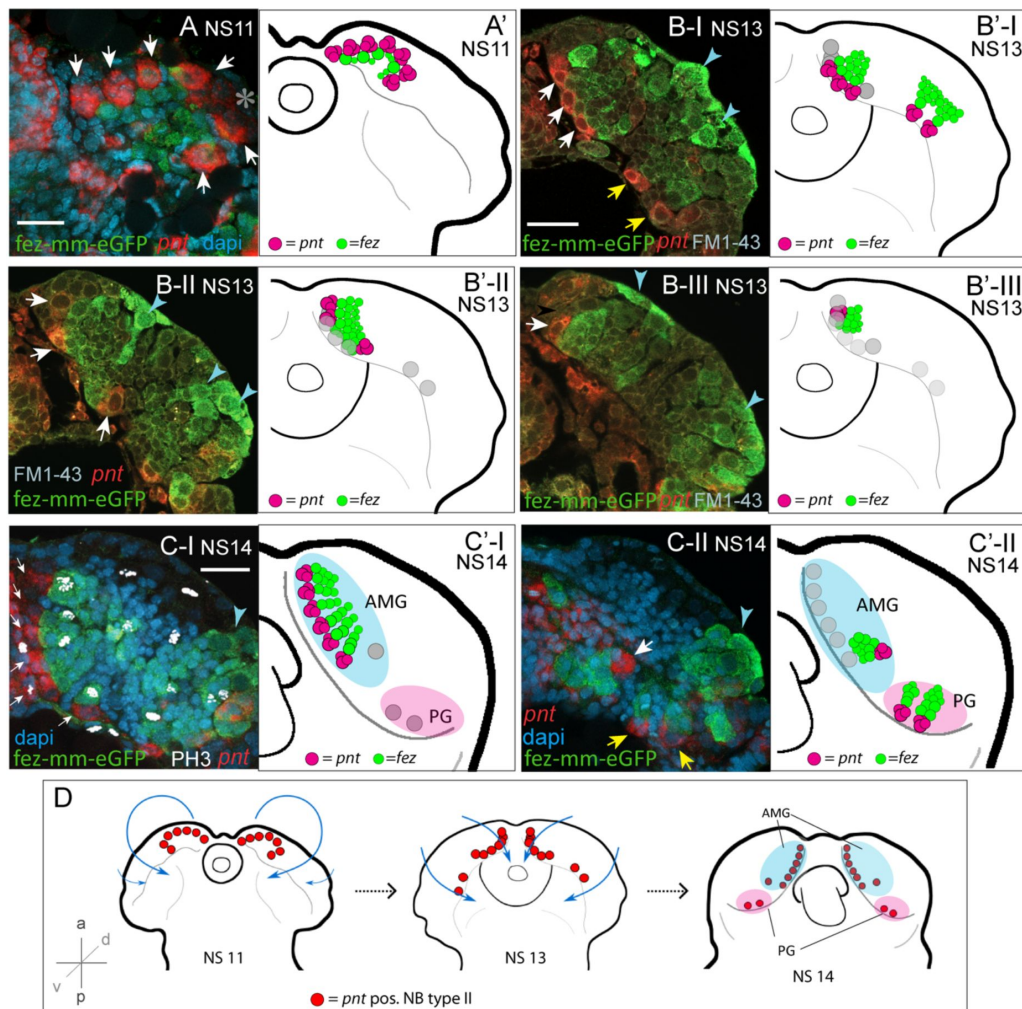


Figure 3

Developmental occurrence and progression of *Tc-pnt*-clusters and associated *fez-mm-eGFP* cells (type-II NB lineages).

A-C) Anti-GFP antibody staining reflecting *fez-mm-eGFP* expression in combination with *Tc-pnt* RNA *in situ* hybridisation and dapi labelling of nuclei (A, C) or FM1-43 membrane staining (B), single planes from confocal z-stacks of right head lobes. Scalebars are 25 μ m. A'-C') Schematic drawing of the head lobe at the respective stages; clusters out of focus depicted by grey circles. D) Schematic depiction of the localization of type-II NBs during *Tribolium* head morphogenesis. **A)** At stage NS11 seven *Tc-pnt*-clusters (white arrows) are found in an anterior-medial horseshoe-like arrangement with *fez-mm-eGFP* cells in the centre. Asterisk indicates the position of one *Tc-pnt*-cluster which is outside the focal plane of this image. **B)** At stage NS13 the seven anterior-medial clusters have been moved to the medial margin of the developing brain by morphogenetic movements of the head lobes (white arrows). The clusters are situated in different dorso-ventral levels shown in the different images B-I to B-III. Two additional posterior *Tc-pnt*-clusters have appeared (yellow arrows). Blue arrows point to *fez-mm-eGFP* positive cells that are not part of the type-II NB lineages. **B-I:** ventral level **B-II:** mid-level, **B-III:** dorsal level. **C)** At stage NS14 a group of six clusters are arranged in one plane anterior-medially in the brain lobe (white arrows) (**C-I**). Anti-PH3 labelling marks mitotic cells within the lineages (white signal) and clusters out of the focal plane are indicated by grey circles in C'-I. **C-II** One cluster of the anterior median group (white arrow) is in a deeper plane, as well as the two posterior clusters (yellow arrows). Blue arrowheads point to lateral *fez-mm-eGFP* expression not associated with type-II NBs. A'-C') AMG =anterior medial group (blue), PG =posterior group (pink). Grey circles in B'-C' indicate position of type-II NBs in different focal planes. **D)** *Tc-pnt*-expressing cells identified as type-II NBs first appear bilaterally in the anterior most part of the embryonic head (stage NS11). During head morphogenesis this anterior tissue folds over so that the type-II NBs end up in a more medial position. Two more type-II NBs emerge posteriorly (stage 13). At the end of embryogenesis (stage NS14) the type-II NBs can be divided in an anterior medial group (AMG (blue); including type-II NBs 1-7, numbering starting from anterior) and a posterior group (PG (pink), including type-II NBs 8-9).

medial border of the head lobes and their number has increased to nine. The seven clusters that were already present at stage NS11 are arranged in an anterior group (white arrowheads in **Fig. 3B**) at stage NS13 while two additional clusters have emerged more posteriorly (yellow arrows in **Fig. 3B-I**). The clusters are in different depths (reflected by the three optical sections shown in **Fig. 3 B-I** to **B-III**, clusters out of focus are visualized by grey circles in **Fig. 3 B'I-II**, respectively). Fez-mm-eGFP expression not related to type-II NB offspring was found in the lateral head lobe from NS13 onwards (blue arrowheads in **Fig. 3C**). At stage NS14 six of the anterior-medial clusters are aligned in one plane along the medial rim of the developing brain (**Fig. 3 C-I**), whereas the most posterior cluster of the anterior group is located at a separate position and in a deeper plane (**Fig. 3C-II**). The posterior group is found at about the same focal plane as well (**Fig. 3 C-II**). The aligned six clusters produce their *Tc-fez/erm*-positive offspring towards laterally whereas the *Tc-fez/erm*-positive offspring of the more posterior cluster is found medially of it (**Fig. 3 C-II**). The two most posterior clusters produce *Tc-fez/erm*-cells in an anterior-lateral direction (**Fig. 3 C-II**). The apparent re-arrangement of the *Tc-pnt*-clusters from the anterior rim of the embryonic head to a medial position reflects not active migration but follows the overall tissue movements during head morphogenesis that was previously described in *Tribolium* [19] (**Fig. 3 D**). Based on their position by the end of embryogenesis (stage NS14) we assign the nine type-II NB clusters into two groups: one anterior median group consisting of seven type-II NBs and one posterior group consisting of two type-II NBs (see **Fig. 3 C** and **D**).

Characterization of cell- and nuclear size of *pnt*-positive type II NBs

We wanted to confirm that each of the *Tc-pnt* and *Tc-fez/erm* expressing lineages contains a neuroblast. Neuroblasts, including type-II NBs, are larger than other cells, have larger nuclei and are mitotically active [40]. Therefore, we determined the cell size of the largest cells of the *Tc-pnt* cluster and compared it to a random sample of cells of the embryonic head. Both samples were taken from stage NS13 and NS14 embryos. We found that the cells that we had assigned as Type-II NBs had a significantly larger diameter (av.: 12.98 μ m; n=43) than cells of the reference sample (av.: 6.75 μ m; n=1579) (**Fig. 4**). We also found that the diameters of nuclei of type-II NBs were significantly larger (av.: 8.73 μ m; n=17) than the ones of a control sample (av.: 5.21 μ m; n=1080) (**Fig. 4**). Although type-II NBs were clearly larger than the average cell of the head lobes the control sample also contained cells of an equal size (**figure 4**) which may constitute type-I NBs.

Conserved patterns of gene expression mark *Tribolium* type-II NBs, different stages of INPs and GMCs

Drosophila type-II NBs, INPs and GMCs express a specific sequence of genes including *asense* (*ase*), *deadpan* (*dpn*) and *prospero* (*pros*) (**Fig. 1**) [30]. However, these markers have up to now not been tested for expression in type-II lineages in other organisms. We tested if the respective lineages express these markers in a comparable sequence in the beetle and if they mark distinctive cell types within the lineages (**Fig. 5 panel A-D** and **Fig. panel 6 A, B**). We found that the large *Tc-pnt*-positive but *fez-mm-eGFP* negative cells (i.e. type-II NBs) express the gene *Tc-dpn*, in line with expression of this gene in *Drosophila* neural precursors (**Fig. 5, panel A**). Like *Drosophila* type-II NBs, these cells do not express the type-I NB marker *Tc-ase* (yellow arrowhead in **Fig. 5, panel D**). We further found additional cells directly adjacent of the type-II NBs itself which we believe are recently born immature INPs. They are *Tc-pnt*-positive but *fez-mm-eGFP*-negative and neither express *Tc-ase* (**Fig. panel D**, pink arrowheads). Expression of *Tc-ase* starts in the following in immature *Tc-pnt*-positive INPs (**Fig. 5 panel D**; blue arrowhead) and is maintained in INPs that are *Tc-pnt*-negative but are marked by *fez-mm-eGFP* (**Fig. 5, panel D** and **E**). *Tc-dpn* is absent from immature *Tc-pnt*-positive INPs (**Fig. 5D**, pink arrowhead) but is expressed in mature INPs (orange arrowheads in **Fig. 5, panel B**). Cells that are located at the distal end of the lineages that do not express *Tc-dpn* but are positive for *fez-mm-eGFP* and *Tc-ase*

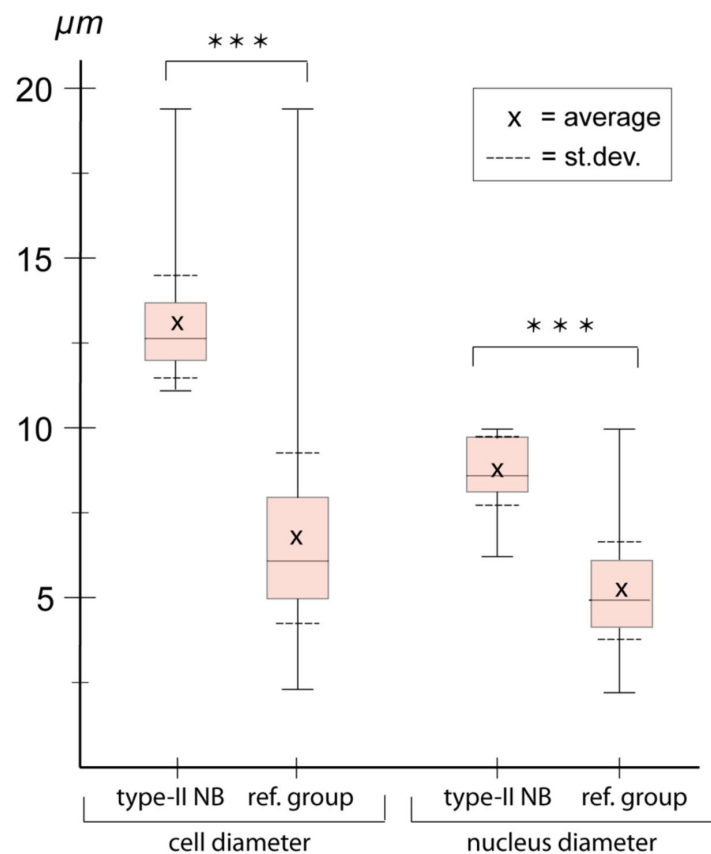


Figure 4

Cell- and nuclear diameter of type-II NBs compared to a control group.

Stages NS13-14. Left: diameter of *Tc-pnt*-pos. type-II NBs (average diameter=12.98 μm ; n=43) compared to control group (average diameter=6.75, n=1579). Right: average nuclear diameter of type-II NBs (=8.73 μm , n=17) compared to average nuclear diameter of control group (=5.21, n=1080). Differences between groups are significant (t-test; *** $\equiv P < 0.001$).

are classified as GMCs (**Fig. 5, panels B, C** and **E**). We found the gene *Tc-pros* expressed in most *fez-mm-eGFP* expressing cells of a lineage (in **Fig. 6, panel A-B**) and infer from the extent of the expression that it is expressed in mature INPs (*Tc-dpn+*) and GMCs (*Tc-dpn-*). *Fez-mm-eGFP* positive cells at the base of the lineage that do not express *Tc-pros* are immature INPs (**Fig. 6, panel A, B**; blue arrowheads). In summary, the expression dynamics found in the *Tribolium* type-II NB lineages are very similar to the one found in *Drosophila*. Therefore these neural markers can be used for a classification of cell types within the lineages into type-II NBs (*Tc-pnt+*, *Tc-fez/erm-*, *Tc-ase-*, *Tc-dpn+*, *Tc-pros-*), immature-I INPs (*Tc-pnt+*, *Tc-fez/erm-*, *Tc-ase-*, *Tc-dpn-*, *Tc-pros-*), immature-II INPs (*Tc-pnt+*, *Tc-fez/erm+*, *Tc-ase+*, *Tc-dpn-*, *Tc-pros-*), mature INPs (*Tc-pnt-*, *Tc-fez/erm+*, *Tc-ase+*, *Tc-dpn+*, *Tc-pros+*), and GMCs (*Tc-pnt-*, *Tc-fez/erm+*, *Tc-ase+*, *Tc-dpn-*, *Tc-pros+*). This classification is summarized in **Fig. 7 A-B**. Image stacks from specimens co-stained with the anti-PH3 mitosis marker (data shown in **fig. 3C** and listed in **table 1** and **2**, available at <https://doi.org/10.25625/8IVICL>) indicate that immature-I and -II INPs are not dividing whereas mature INPs are.

Type-II NB lineages produce central complex cells that are marked by shaking hands (*skh*)

To test if the *Tribolium* embryonic type-II NB lineages contribute to the beetle central complex like in the fly, we used the shaking hands- (*skh*-) reporter line that marks central complex neurons and their postmitotic embryonic precursors [41]. By co-staining for *Tc-pnt* and *Tc-fez/erm* RNA in the background of the *skh*-line we found that the *skh*-positive cells are located directly distally of the type-II NB lineages in a position where the type-II NBs derived neural cells are expected (**Fig. 8 panels A-B, E,F**). Most *skh*-cells have switched off *fez/erm*-expression, but at the transition between *Tc-fez/erm* and *skh* cells we found some cells that express both markers (**Fig. 8 panel D, E**), supporting the view that many *skh*-positive central complex cells stem from the *fez*-expressing GMCs of the type-II NB lineages (**Fig. 8 E**). We found *skh* cells at the end of lineages of the anterior-median group (**Fig. 8, panel A**) and at the end the two posterior lineages (**Fig. 8, panel B**). We therefore conclude that both groups contribute to central complex neuropile (**Fig. 8 F**). We cannot say whether the lineage of type-II NB seven, which we assigned to the anterior-median group, but which is located with some distance to the other type-II NB of that group (**Fig. 3 C, D**; grey circle in **Fig. 8 F**), produces *skh+* cells, but we assume that it does.

The *Tribolium* embryonic lineages of type-II NBs are larger and contain more mature INPs than those of *Drosophila*

In beetles, a single-unit central body (most likely constituting the fan-shaped body [10]) develops during embryogenesis and has gained functionality at the onset of larval life [15]. By contrast, in *Drosophila* central complex commissural tracts originating from type-II NBs form in the embryo but the type-II NB become quiescent and only resume divisions in the late larval stage [29,50]. In stark contrast to *Tribolium* a functional central complex is only observed in the adult fly [3]. In addition, the volume of the L1 central brain including the central complex is about four times larger in *Tribolium* than it is in *Drosophila* [53], while there are only small differences in the relative neuropilar volumes of the adult beetle and fly central bodies [53,54]. As type-II NBs contribute to central complex development we asked in how far the embryonic division pattern of these lineages in the beetle would reflect this heterochronic development. To assess the size of the embryonic type-II NB lineages in beetles we counted the *Tc-fez/erm* positive (*fez-mm-eGFP*) cells (INPs and GMCs) of the anterior medial group (type-II NB lineages 1-7). As we were not always able to distinguish between cells belonging to neighbouring INP-lineages we quantified and averaged the cell number of all lineages of the anterior medial group. We also evaluated the proportion of dividing cells in that group based on anti-PH3 staining (**table 1**).

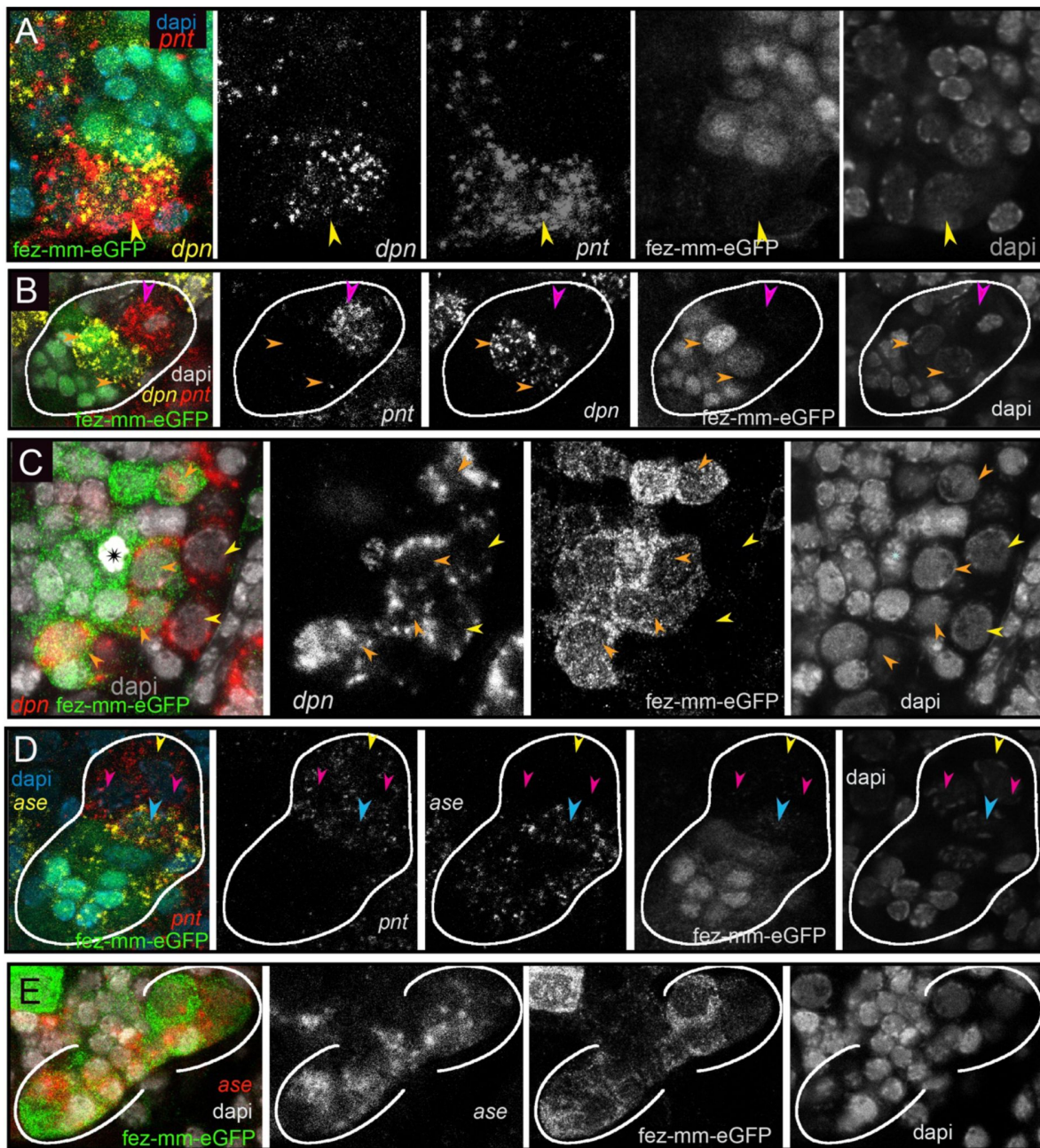


Figure 5

Differential expression of the neural markers *Tc-pnt*, *Tc-fez/erm*, *Tc-dpn* and *Tc-ase* in type-II NBs, INPs and GMCs.

A, B, D) Anti-GFP staining visualizing *fez-mm-eGFP* expression in combination with HCR-labelling of further factors. C, E) GFP antibody staining in combination with RNA *in situ* hybridisation. A-E) Dapi staining of nuclei. All panels are single planes from confocal z-stacks. **Panel A)** Stage NS12, expression of *Tc-pnt* and *Tc-dpn* in type-II NBs (yellow arrowhead) with adjacent *fez-mm-eGFP* positive INPs. **Panel B)** Stage NS13, white line marks one lineage. Expression of *Tc-dpn* is absent in the immature-I *Tc-pnt*+ INPs (pink arrowhead) but present in mature INPs (orange arrowheads), white line marks one lineage. Type-II NB not in focus of this image. **Panel C)** Stage NS14, expression of *Tc-dpn* in type-II NBs (yellow arrowheads), and mature INPs (orange arrowheads). Asterisk marks mitosis in a *fez-mm-eGFP* cell visible through the condensation of chromatin stained by dapi. **Panel D)** Stage NS13, *Tc-ase*- and *Tc-pnt*-expression in type-II NBs and INPs, white line marks one lineage. Yellow arrowhead: type-II NBs (*Tc-pnt*+), pink arrowheads: immature-I INP (*Tc-pnt*+), blue arrowhead: immature-II INP (*Tc-fez*+/*Tc-ase*+/*Tc-pnt*+). **Panel E)** Stage NS13, expression of *Tc-ase* in *fez-mm-eGFP* positive cells, location of type-II NBs and immature-I INPs is outside the focus.

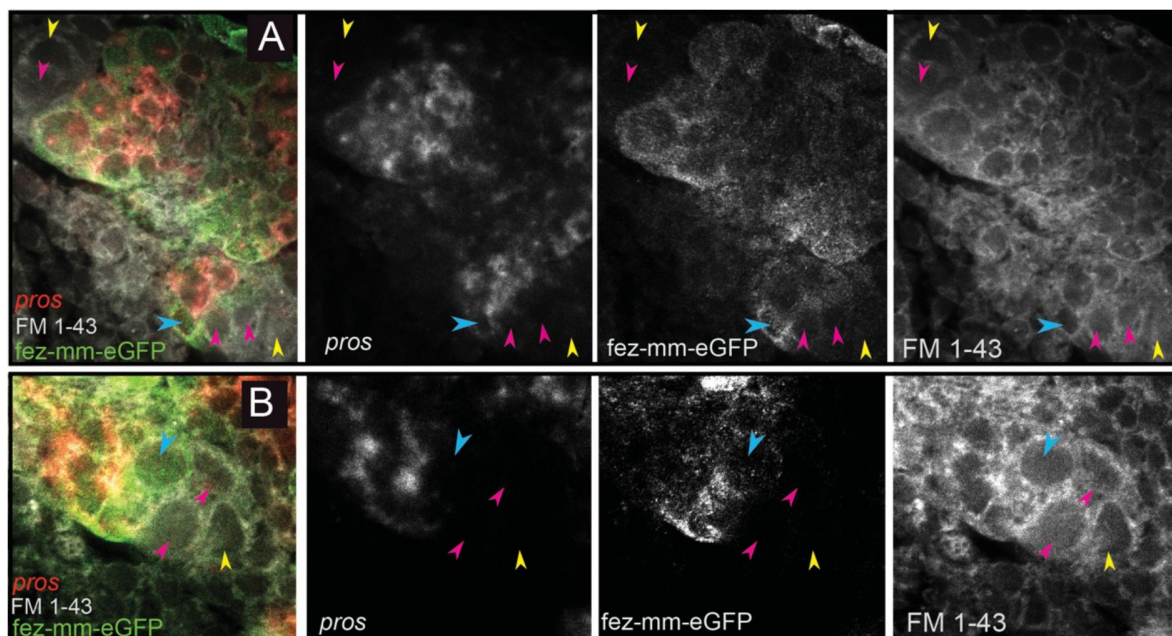


Figure 6

Expression of *Tc-pros* in *fez-mm-eGFP* cells.

A-B) Anti-GFP antibody staining in combination with *Tc-pros* RNA *in situ* hybridization. Membranes stained with FM 1-43. Single planes from confocal z-stacks. **Panel A)** Stage NS13, two type-II NB lineages producing INPs and GMCs in opposing directions. Expression of *Tc-pros* in mature INPs and GMCs. Cluster of type-II NBs (yellow arrowheads) and immature-I INPs (pink arrowheads). Immature-II INPs express *Tc-fez/erm* but not *Tc-pros* (blue arrowhead). **Panel B)** Stage NS13, detail of *Tc-pros*-negative type-II NBs (yellow arrowhead) and immature-I INPs (pink arrowheads) and immature-II INP (blue arrowhead). Note that distinction between type-II NB and immature INPs in **A)** and **B)** is only made based on position.

Figure 7

Schematic drawing of expression of different markers in a type-II NB lineage.

A) and B) show the same lineage with *Tc-fez/erm* and *Tc-pnt* expression but with different additional markers mapped on top. **A)** Subclassification into type-II NBs (*Tc-pnt*+), immature-I INPs (*Tc-pnt*+), immature-II INPs (*Tc-pnt*+, *Tc-fez/erm*+, *Tc-ase*+) and mature INPs and GMCs (*Tc-fez/erm*+, *Tc-ase*+) (based on stainings shown in Fig. 5). **B)** Type-II NB (*Tc-pnt*+, *Tc-dpn*), immature-I (*Tc-pnt*+) and immature-II (*Tc-pnt*+, *Tc-fez/erm*+) . Mature INPs express *Tc-dpn*, *Tc-pros* and *Tc-fez/erm* whereas GMCs express only *Tc-fez/erm* and *Tc-pros* (based on staining shown in Fig. 5 and 6). Note that of the selected markers immature-I INPs express *Tc-pnt* only.

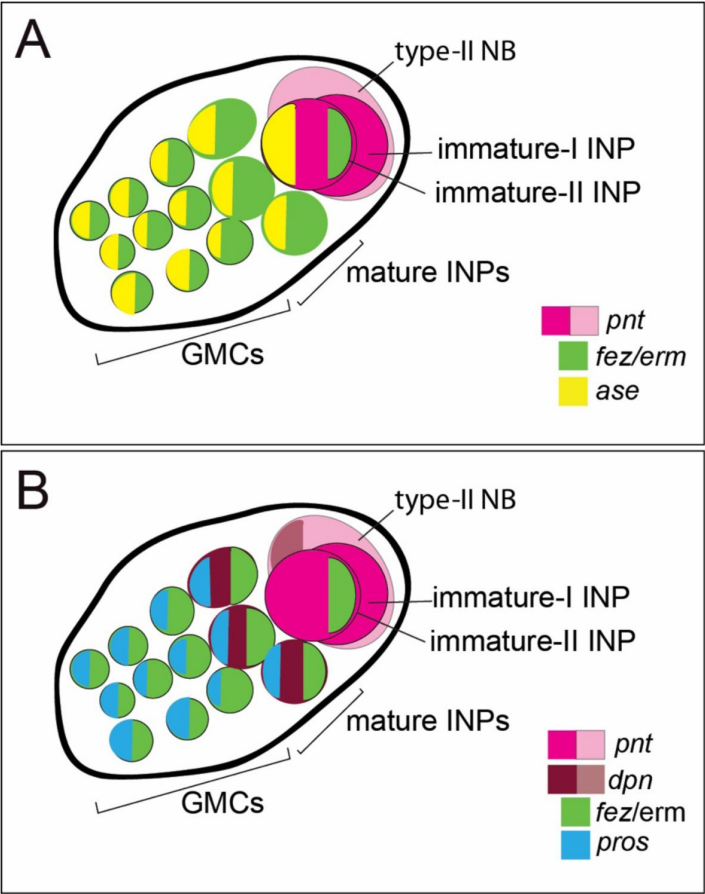


Table 1

Tc-fez/erm positive cell number (type-II NBs, INPs and GMCs)

Embryo#/stage	Total number of cells of 7 anterior lineages	% dividing cells (PH3 antibody)	Cells per lineage (average)
1/NS 13	146	8.2%	20.9
2/NS 13	111	6.3%	15.9
3/NS 14	139	5.0%	19.9
4/NS 14	96	10.5%	13.7

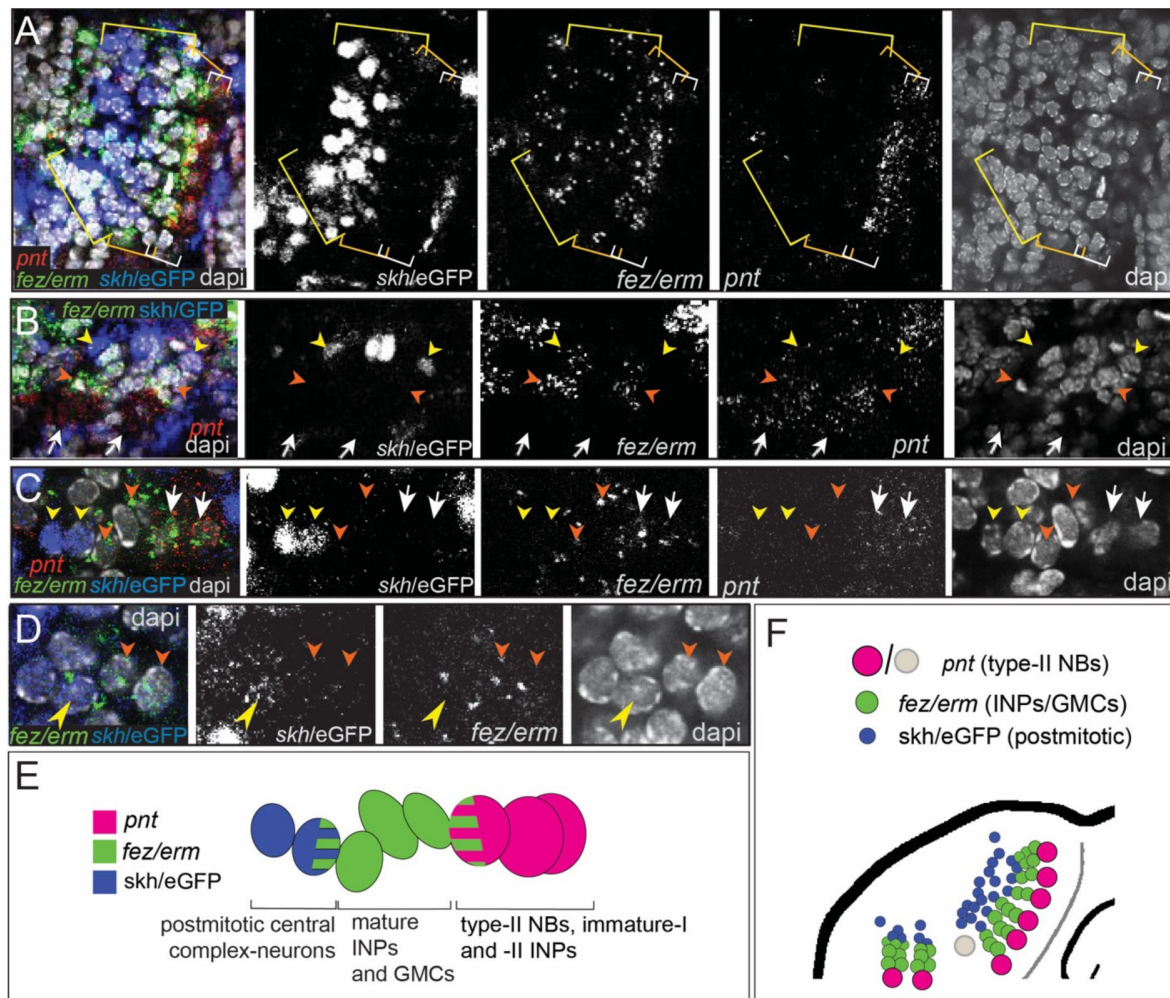


Figure 8

Expression of the transgenic central complex reporter skh/eGFP in relation to type-II NB lineages.

A-D) Antibody labelling of eGFP, which is expressed in the pattern of *skh* [41] in combination with HCR visualizing *Tc-fez/erm* and *Tc-pnt*, dapi labelling of nuclei, single planes of confocal z-stacks, all stage NS14. **A**) Left head lobe, anterior group of *Tc-pnt*-positive type-II NBs, *Tc-fez/erm*-expressing INPs/GMCs and *skh/eGFP* positive postmitotic cells. **B**) Left head lobe, posterior group of type-II NB lineages. *Tc-pnt*⁺ type-II NBs or immature-I/-II INPs (white arrows), *Tc-fez/erm*⁺ INPs/GMCs (orange arrowheads) and *skh/eGFP* postmitotic cells (yellow arrowheads). **C**) Detail of one lineage including type-II NB cluster (*Tc-pnt*⁺, white arrows), INPs/GMCs (*fez*⁺, orange arrowheads) and *skh/eGFP* cells at the end of the lineage (yellow arrowheads). **D**) Detail of transition from *Tc-fez/erm*⁺ cells (orange arrowheads) to *skh/eGFP* cells reveals a small area of co-expression of both factors (yellow arrowhead). **E**) Schematic drawing of type-II NB lineage showing relative positions and markers of type-II NBs, INPs/GMCs and postmitotic central complex-forming cells. **F**) Schematic overview of the arrangement of the type-II NB lineages including *skh/eGFP*⁺ central complex-forming cells in the head lobe of stage NS14 (immature-I/-II INPs not shown for simplicity). No data on *skh*-GFP expression in lineage of type-II NB 7 (marked in grey) is available.

We found that at stage NS13 (early retracting embryo) each lineage consisted on average of 18.4 progenitor cells and at stage NS14 (retracted embryo) of 16.8 progenitors. Between 5-10 % of the cells were marked by the PH3 antibody. As PH3 marks dividing cells in a specific phase, the portion of dividing cells may be higher. We further quantified the number of mature *Tc-dpn*-positive INPs (**table 2** [↗](#)).

We used the numbers from *Tribolium* (**table 1** [↗](#), 2) for comparison with type-II NB lineages from *Drosophila* embryos (data from [\[29\]](#) [↗](#)). At fly embryonic stage 13, which is comparable to *Tribolium* stage NS14 in that it represents a retracted germband, lineages only contained one or more INPs but no GMCs [\[29\]](#) [↗](#). Therefore, we compared our data for the anterior lineages of *Tribolium* stages NS13 and NS14 to the anterior and median cluster in the *Drosophila* stage 15 and 16 embryos. In both species these stages are the two penultimate stages of embryogenesis when all type-II NB lineages are present, and importantly, in *Drosophila* the maximum lineage sizes and numbers of INPs are reached at these stages. We found that the beetle lineages were significantly larger than the corresponding ones in *Drosophila* (**Fig. 9** [↗](#); *Drosophila* data taken from [\[29\]](#) [↗](#)). Of note, cell counts in *Drosophila* also included neurons [\[29\]](#) [↗](#). Our counts are based on *Tc-pnt* and *fez-mm-eGFP* expression and therefore mostly included progenitors. However, based on the small area of co-expression of *Tc-fez/erm* and *skh*-expression (see above/ **figure 8** [↗](#)) they may also include some newly born neurons, but the counts in *Drosophila* included more mature neurons which further increases the difference in the number of progenitor cells between the two species and the observed difference in lineage size represents a minimum estimate.

Due to a lower quality of the image data from the posterior group which is in a deeper layer (see **Fig. 3** [↗](#)) we were not able to quantify cells of this cluster in *Tribolium*. In *Drosophila*, the posterior lineages are larger (av. 10 cells [\[29\]](#) [↗](#)) than the anterior and median ones, but they are still smaller than the anterior lineages of *Tribolium* at the stages that were investigated. We also found that the embryonic type-II NB lineages in *Tribolium* comprised significantly more mature INPs (*Tc-dpn*+/*Tc-fez*+) than the corresponding lineages in *Drosophila* (**Fig. 9** [↗](#)). The described observations on the higher numbers of progenitor cells in *Tribolium* when compared to *Drosophila* correlate with a previously described earlier (i.e. embryonic) formation of a functional central complex in beetles, which is present in the first larval stage [\[3\]](#) [↗](#).

Type-II NBs and their lineages are differentially marked by the head patterning transcription factors *Tc-six4* and *Tc-six3* but do not express *Tc-otd*

The anterior-most embryonic insect head is divided into different territories by highly conserved developmental transcription factors that give rise to different parts of the protocerebrum [\[11\]](#) [↗](#). Hence, we wondered in which of these molecular territories subsets of type-II NB lineages are located, as neuroectoderm regionalization genes are candidates for giving specific spatial identities to individual lineages. We tested co-expression with the previously characterized head patterning transcription factors *Tc-otd*, *Tc-six4*, *Tc-six3*, known to be active in the developing protocerebrum (*Tc-otd*) and in the anterior medial region of the neuroectoderm, where type-II NBs emerge (*Tc-six3*, *Tc-six4*) [\[11\]](#) [↗](#), [\[19\]](#) [↗](#), [\[55\]](#) [↗](#), [\[56\]](#) [↗](#). We found no co-expression of *fez-mm-eGFP* marked cells of both the anterior medial and posterior lineages with *Tc-otd*. Rather, *Tc-otd* was expressed in the surrounding embryonic head tissue. *Tc-otd* was also absent from the type-II NBs themselves (**Fig. 10 panel A-B** [↗](#)). However, *Tc-otd* might be expressed in neural cells derived from these lineages as we can detect expression in the central area of the brain lobes where for instance *skh* cells are located (**Fig. 8 panel A, F** [↗](#); **Fig. 10 panel A-B** [↗](#)). Interestingly, we found that *Tc-six4* is expressed specifically in that area of the head lobe in which the type-II NBs clusters 1-6 of the anterior medial group first differentiate. At stage NS13 *Tc-six4* marks both, type-II NBs and *fez-mm-eGFP* expressing INPs and some surrounding cells (**Fig. 11, panel A** [↗](#)). At the following differentiation stage (NS14) *Tc-six4* still marks the type-II NBs clusters of the anterior medial group but is not expressed in the *fez-mm-eGFP* INPs (**Fig. 11 panel B** [↗](#)). It is also not

Table 2

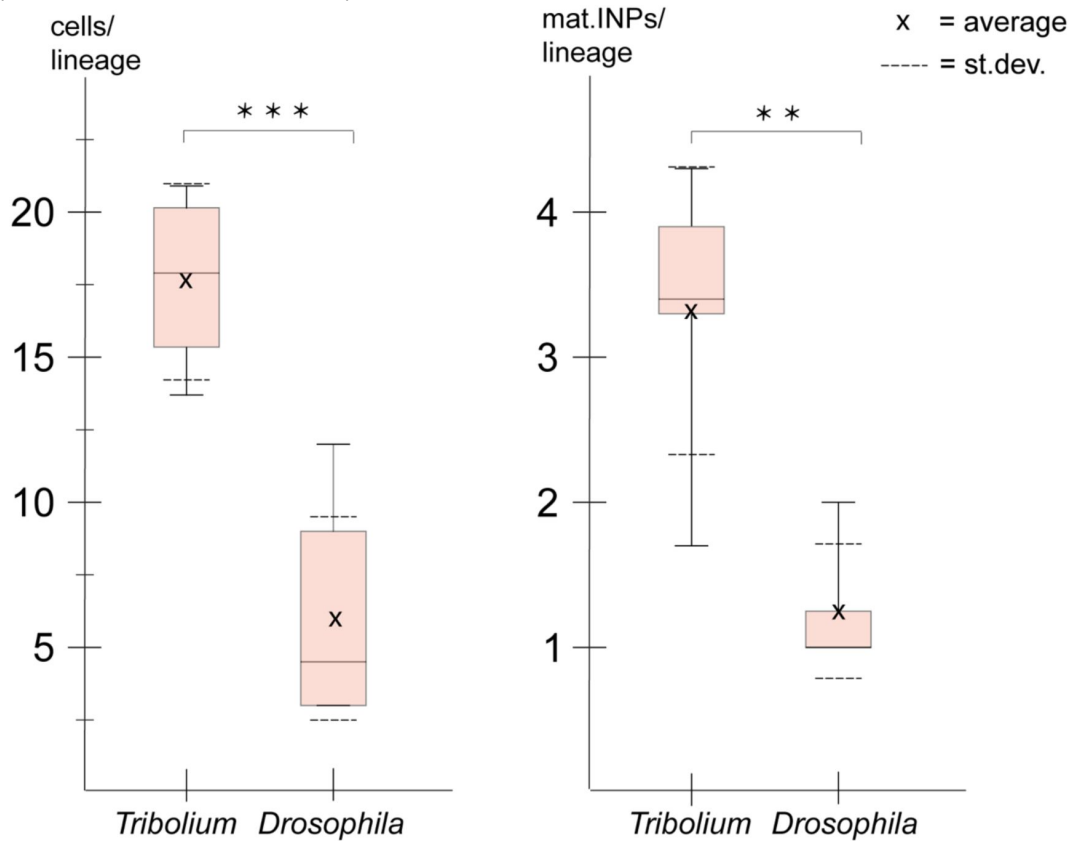
Mature INPs (*Tc-fez/erm* and *Tc-dpn* positive)

Embryo #/ stage	Total number of mature INPs of 7 anterior lineages	% dividing cells (PH3 antibody)	Number of mature INPs per lineage (average)
1/NS 13	24	n.d.	3,4
2/NS 13	12	n.d.	1,7
3/NS 13	23	n.d.	3,3
4/NS 14	27	3,7%	3,9
5/NS 14	30	13,3%	4,3

Figure 9

Size comparison of *Tribolium* and *Drosophila* type-II NB lineages.

Anterior/median lineages from *Tribolium* stages 13 and 14 (quantified in this work, see [table 1](#) and [2](#)) compared to *Drosophila* stages 15 and 16 (pooled data from [\[29\]](#)). Left: Lineage sizes comprising type-II NBs, INPs and GMCs (*Tribolium* n=4; *Drosophila* n=8). Right: Number of *Tc-dpn*-positive mature INPs (*Tribolium* n=5; *Drosophila* n=8). T-test p-value significance level (t-test; *** $\equiv$ $P < 0.001$, ** $\equiv$ $P < 0.01$).



expressed in the posterior group of type-II NBs but only marks type-II NBs of the anterior medial group. Finally, we found that *Tc-six3* marks, and is restricted to, the anterior-most lineages 1-4 including both the type-II NBs and the *fez-mm-eGFP* marked INPs/GMCs (**Fig. 11 panel C**). These results reveal candidates for distinguishing type-II identity from type-I NBs in the brain. Only the latter is possibly marked by *Tc-otd* as *Drosophila otd* is expressed in type-I brain neuroblasts [57]. By contrast, *Tc-six4* only marks type-II NBs and appears to be an early determinant of the anterior medial group. *Tc-six4* is also a candidate for distinguishing the anterior medial from the posterior cluster. The results also reveal a subdivision of the anterior cluster by *Tc-six3*, which marks the four most anterior type-II NBs.

Discussion

Evolutionary divergence of number and grouping of embryonic type-II NB lineages between beetle, fly and grasshopper

We have identified a total of 9 type-II NB lineages on each side of the *Tribolium* embryonic head. This represents a major evolutionary divergence as in both, the hemimetabolan grasshopper *Schistocerca gregaria* and the fruit fly *Drosophila melanogaster* only 8 embryonic type-II NBs were identified [29,38]. Our finding of an additional type-II NB lineage at the embryonic stage shows that the number of type-II NBs is not fixed across insects and suggests that evolutionary divergence in brain development between species may be driven by differences in the number of progenitor lineages. To test this, further research is required to determine where the additional lineage in *Tribolium* is located and if it is associated with morphological differences of the brain.

Although in the frame of this work we did not perform lineage tracing we can speculate about which NB might be lacking in fly embryos based on our data. In the hemimetabolan *Schistocerca* there is no obvious clustering of type-II NBs, and they are all arranged in one row at the medial rim of the head lobes. In *Tribolium* we have observed an arrangement of the type-II NBs into two groups per side, one large anterior medial group containing seven type-II NB clusters and one posterior group of two type-II NBs. The anterior six of the anterior medial group are arranged in one row like in the grasshopper. In *Drosophila* there are three described groups referred to as the anterior, the middle and the posterior cluster [29]. The posterior cluster consists of two type-II NBs and therefore most likely corresponds to the posterior two type-II NBs in *Tribolium*. The *Drosophila* middle and anterior cluster are most likely equivalent to the anterior medial group of *Tribolium*. However, the type-II NB 7, which we assigned to the anterior medial group, but which is a bit separated and produces offspring into the opposite direction (see **Fig. 2 C-II**, white arrow) might be the one that does not have a homologue in the fly embryo. The identification of more specific spatial markers for type-II NBs (in addition to *Tc-six3* and *Tc-six4*) or lineage tracing tools are required to identify the neuroblast, which is not present in fly embryos. Subsequently, it would be especially interesting to see what the role of the additional type-II NBs in the beetle is and which neuropile it contributes to.

Summing up, there is a tendency towards grouping of type-II NBs into subsets in the holometabolan models that was not observed in the only hemimetabolan studied so far. A ninth type-II NB was only found in the *Tribolium* embryo represents a striking developmentally divergent feature, but its role and the homologisation of individual type-II NBs between the different insects requires further studies.

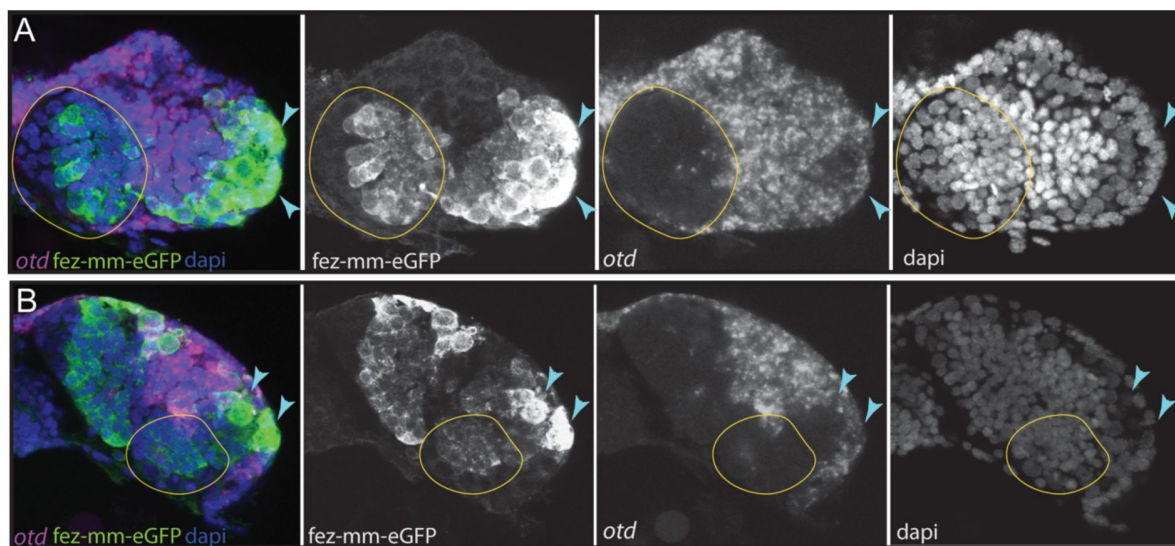


Figure 10

Expression of *Tc-otd* in relation to the *fez-mm-eGFP* lineages.

A/B) RNA *in situ* hybridisation for *Tc-otd* in combination with GFP antibody staining and dapi labelling of nuclei, A and B show projections of different levels of NS14 embryos, right head lobe. **A)** *Tc-otd* is neither expressed in type-II NBs nor the in the *fez-mm-eGFP* positive INPs of the anterior medial group (both positioned within encircled area) but it is expressed in the directly surrounding tissue. **B)** Likewise, *Tc-otd* is not expressed in type-II NBs or *fez-mm-eGFP* cells of the posterior group (encircled area). A lateral area of *Tc-fez/erm* and *Tc-otd* co-expression (**A/B**, blue arrow) likely is part of the eye anlagen [19].

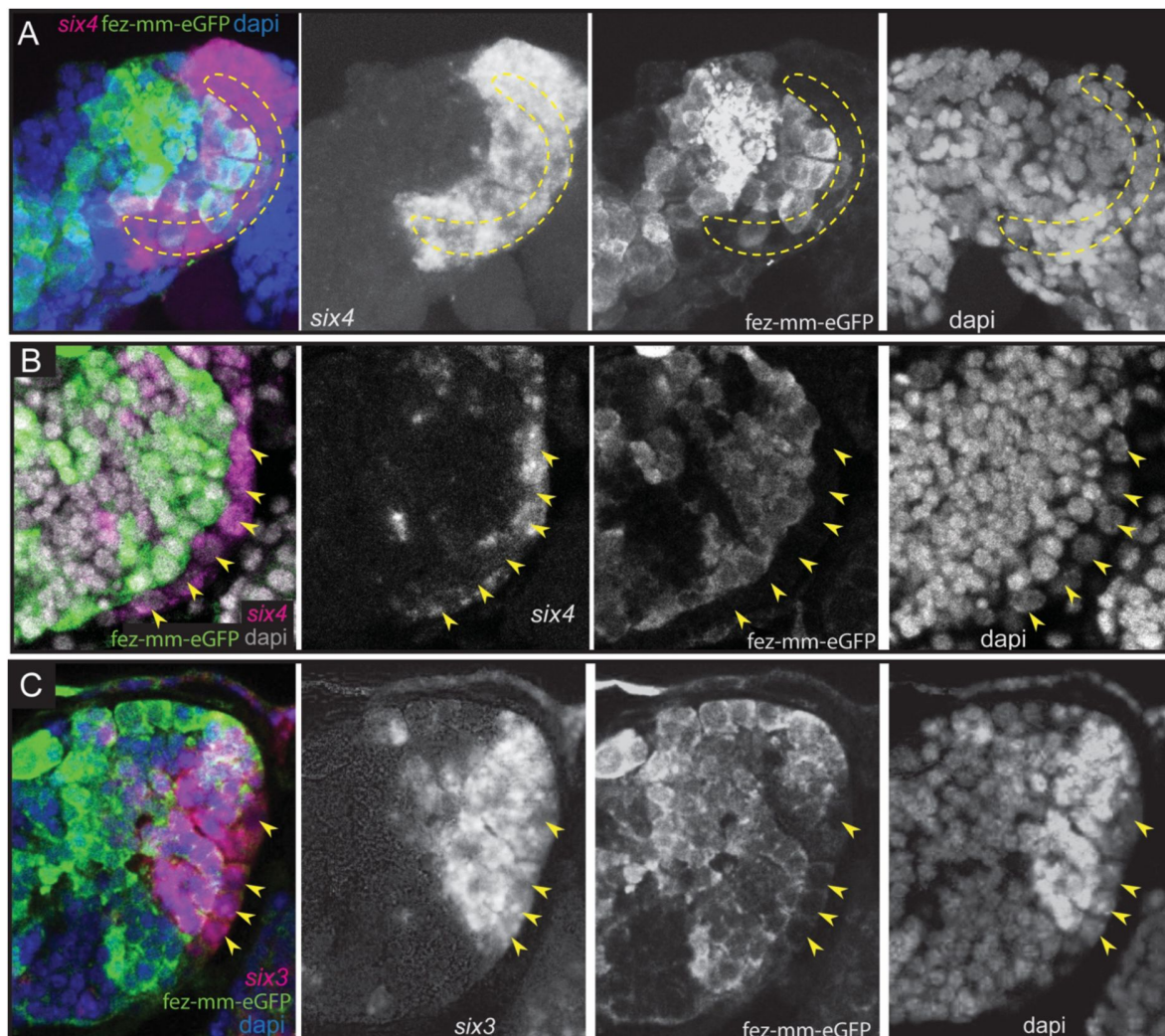


Figure 11

Expression of *Tc-six4* and *Tc-six3* in type-II NBs and INPs.

A-C) RNA *in situ* hybridisations of *Tc-six4* and *Tc-six3* in combination with GFP antibody staining reflecting fez-mm-eGFP expression, dapi labelling of nuclei, left head lobes shown. **Panel A)** Stage NS13; *Tc-six4* is expressed in an area encompassing type-II NBs (positioned in encircled area) and mm-eGFP expressing INPs. **Panel B)** Stage NS14; *Tc-six4* is now restricted to type-II NBs and their youngest progeny that do not express fez-mm-eGFP (yellow arrowheads). **Panel C)** Stage NS14; *Tc-six3* specifically marks the four anterior-most lineages. It is expressed in the type-II NBs at the base of the lineages (yellow arrowheads) and in the fez-mm-eGFP positive INPs and possibly also in GMCs. It is absent from the remaining lineages of the anterior medial group and is also not seen in the posterior group (not shown).

Gene expression identifies homologous cell types and suggests conservation of gene function between fly and beetle type-II NB lineages

The genes defining the different stages of differentiation in type-II NB lineages have been identified and intensively studied in *Drosophila*, but it had remained unclear, in how far the respective patterns and mechanisms were conserved. Our results show a large degree of conservation of expression and probably also function – at least in holometabola. We identified type-II NBs by their larger than average size [40] and the expression of the signature marker *Tc-pnt* adjacent to a group of *Tc-fez/erm* expressing cells (INPs and GMCs). In *Drosophila* both these factors have key functions in defining the developmental potential of type-II NBs and INPs: Pnt suppresses Ase in type-II NBs and promotes the formation of INPs [45]. Loss of *pnt* expression leads to a de-differentiation of INPs as *fez/erm* is no longer repressed in young INPs [44]. Fez/Erm controls proliferation of INPs by activating Pros and prevents dedifferentiation of INPs into type-II NBs [43]. We show that also in *Tribolium* INPs undergo a maturation process, during which expression of *Tc-pnt* stalls and expression of *Tc-fez/erm* is switched on. Despite both factors characterising different stages of INP maturation they are not completely mutual exclusive and we observed a small window of overlap in immature-II INPs, highly suggestive of a conserved role of Pnt in the activation of *fez/erm* [44] (Fig. 12). *Drosophila* *dpn* is a neural marker [58] expressed in type-II NBs and again in mature INPs, leaving a gap of expression in immature-I and II INPs. *Drosophila* *ase* is repressed in type-II NBs but expressed together with *fez/erm* in INPs and GMCs where its expression overlaps with *pros* [59]. We found identical dynamics of gene expression in *Tribolium* (see Fig. 12) suggesting conservation of gene function within the lineages and a conserved process of INP maturation.

In conclusion, expression of the genes examined in this study in type-II NBs and their lineages is highly conserved in fly and beetle, which is implicit of conserved gene function and a conserved process of central complex formation.

Divergent timing of type-II NB activity and heterochronic development of the central complex

Previous work described a heterochronic shift in central complex formation between fly and beetle [3,10]. In *Tribolium* parts of the central complex develop during embryogenesis. It becomes functional at the onset of the first larval stage, and is presumably required for the more complex movements of the beetle larvae using legs [3]. In both *Drosophila* and *Tribolium* the anterior groups of type-II NBs (anterior medial group of *Tribolium* and anterior and middle cluster of *Drosophila*) are fully present in the last 3rd of embryogenesis, with the beginning of germ band retraction [8,29,60]. However, in the *Drosophila* embryo only a limited number of INPs and GMCs are produced before they enter a resting stage prior to hatching [29]. By contrast, we show that *Tribolium* type-II NB lineages are much larger at the embryonic stage compared to *Drosophila* [29]. They also include more mature, *Tc-dpn* expressing cycling INPs which have the capability of driving the increase in lineage size. We believe that this increased activity may contribute to the embryonic development of a functional central body and protocerebral bridge [3]. This hypothesis does however require further confirmation by lineage tracing experiments and by testing how manipulating central genes acting within the lineages will affect the timing of central complex development. In addition, differences in cell number in the individual central complex neuropiles between *Tribolium* and *Drosophila* have to be assessed, as the higher number of progenitors may also reflect a higher number of neurons present in the adult beetle fan shaped body or may be due to a slower rate of division and a resulting accumulation of progenitor cells. Lastly, we do not yet have any information whether *Tribolium* type-II NBs, or *Tribolium* neuroblasts in general, also enter a stage of quiescence at the end of embryogenesis as they do in *Drosophila* [29] and what role neurons born in the 1st larva play in *Tribolium*.

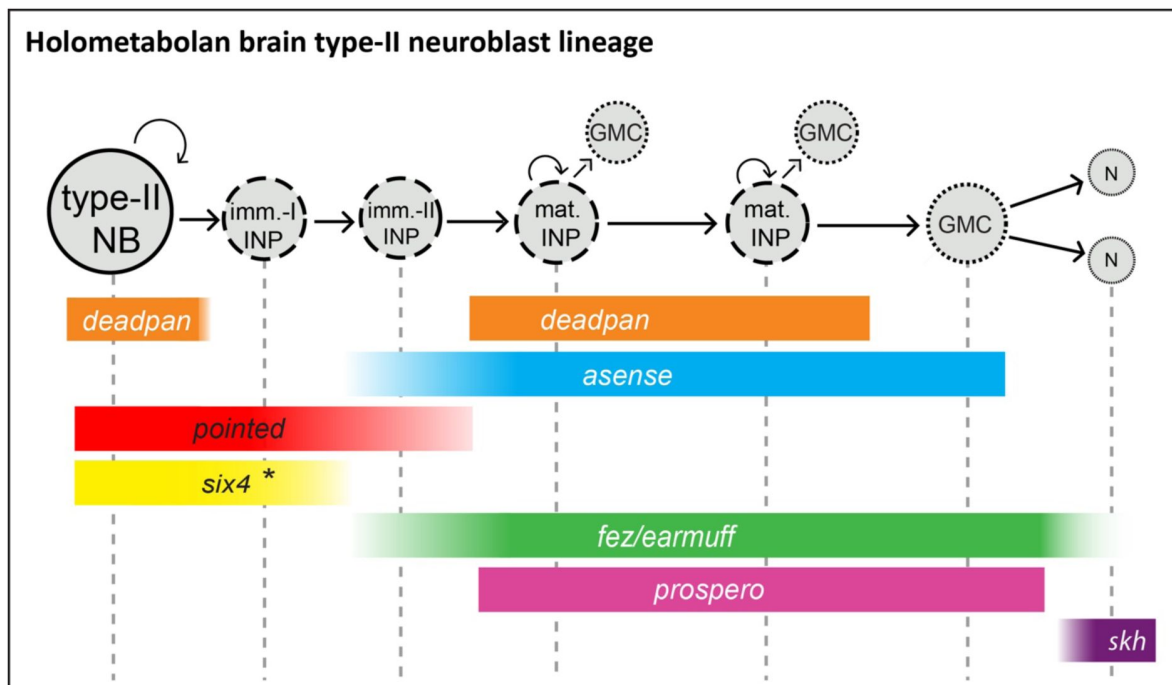


Figure 12

Conserved aspects of gene expression in type-II NB lineages of the holometabolan models *Drosophila* and *Tribolium*.

Overview on the different cell types within one lineage (type-II NBs, INPs, GMCs and central complex neurons), their mitotic activity, and summary of the expression of key factors in the respective cell types. INPs are divided into immature-I, immature-II, and mature INPs with each subtype having a unique expression profile. Note that each lineage can have several mature INPs. We strongly assume that glia are also produced from these lineages in *Tribolium*, as they are in *Drosophila* [30]. * In the *Tribolium* embryo *Tc-six4* is only expressed in the anterior medial lineages, and its expression is overlapping with *Tc-fez/erm* at an earlier stage (NS12) whereas it is restricted to type-II NBs and immature-I INPs, mutually exclusive with *Tc-fez/erm*, at a later embryonic stage (NS14). *Drosophila six4* was found in all 8 larval lineages [47]. The scheme is based on this work and [30, 41, 43, 44, 47]. NB= neuroblast; imm.-I/-II INP = immature-I/-II intermediate progenitor; mat. INP= mature intermediate progenitor; N=neuron; skh=shaking hands (enhancer trap) [41].

Another emerging question is if and when *Tribolium* type-II NB lineages are active during larval development and how that relates to the described steps of central complex development. After preformation in the embryo *Drosophila* type-II NBs have their main period of activity in the 3rd larva and the lineages are much larger and contain more INPs than the *Tribolium* embryonic lineages [30]. Therefore, it would be interesting to see if type-II NB lineages are present and active in the late larval brain of *Tribolium*.

Taken together, we have for the first time identified and molecularly characterized embryonic type-II NBs and INPs as the major central complex progenitor pool in an insect outside *Drosophila*. We have observed quantitative differences within these lineages when compared between *Tribolium* and *Drosophila*. Based on these findings we hypothesize that increased division behaviour of type-II NB lineages is required for the embryonic central complex development as found in beetles. Our findings set the basis for future work on the relationship of the identified type-II NB lineages and the timing of central complex development.

The placodal marker *Tc-six4* may be responsible for the differentiation and the spatial identity of the anterior medial group of type-II NBs

The different type-II NB lineages contribute to different parts of the brain. For instance, only the anterior four type-II NB lineages form the w, x, y, and z tracts of the central complex in flies and grasshoppers. Hence, there must be signals that make them different despite the conserved and well-studied sequence of neural gene expression and functional interactions in *Drosophila* (Fig. 12) [30,43,44,59]. In the ventral nerve cord, the different identities of the NBs are specified by the combination of spatial patterning genes expressed in the neuroectoderm at the time of delamination [24]. However, little is known about these signals for type-II NBs. In *Tribolium* the domains of head patterning genes in the brain neuroectoderm are well characterized and some highly conserved territories like the anterior-medial *six3*-positive domain were described and suggested to give respective NBs different spatial identities [11,19,55]. The *Tc-six4* expression domain is also largely coincident with an embryonic structure termed insect head placode, a neurogenic, invaginating tissue [56,61] potentially homologous to the vertebrate adenohypophyseal placode which is also marked by *six4* [11,56]. A placodal origin of *Drosophila* type-II NBs has been shown for one type-II NB, and it is assumed that the other type-II NBs also stem from placodal tissue [29,50,62].

Tc-six4 is also a very interesting factor with regards to the specification of type-II NBs because it is expressed only in the anterior group of type-II NBs. Hence, it could contribute to distinguishing their developmental fate from the posterior group. This is however different in the *Drosophila* larva. Here *Six4* is expressed in all eight lineages where it prevents the formation of supernumerary type-II NBs and a premature differentiation of INPs [47]. In the *Tribolium* embryo *Six4* may have a similar rate limiting role within the anterior group of type-II NBs. However, its early expression in a small part of the neuroectoderm [56] and the delamination of the anterior-medial type-II NB group within this domain also hints at an instructive role of *Tc-six4* in the formation of the anterior medial group.

Tc-six3 marks a subset of type-II NBs whereas *Tc-otd* is absent from all lineages

Importantly, we found that in *Tribolium* late embryogenesis, *Tc-six3* is expressed specifically within the lineages of type-II NBs 1-4 of the anterior medial group, but not in the other lineages. In grasshopper and fly these anterior 4 lineages (DM1-4) give rise to the z, y, x and w tracts [25,38] and contribute crucially to the development of protocerebral bridge and central body. These tracts form a major midline crossing neuropile of the central body which serves as a scaffold in the development of this structure [3,25]. We conclude that in *Tribolium* the

evolutionary ancient *six3* territory gives rise to the neuropile of the z, y, x and w tracts. This might well be the same in other insects such as flies and grasshoppers but has to our knowledge not been looked at in detail in these models. In the beetle the central body is also missing in weak *Tribolium* *Tc-six3*-RNAi phenotypes, suggesting that *Tribolium* *six3* is required for the formation of this structure [19]. Given that central complex neuropile is marked by additional factors such as *Tc-foxQ2* (*Drosophila* fd102c [63]) and *Tc-rx* [3,17], it would be interesting to see the relationship of additional spatial patterning genes with regards to the type-II NB lineages in future studies.

The gene *otd* is a marker of the posterior protocerebrum and *six3* and *otd* are believed to subdivide the embryonic anterior brain into two major domains [55]. This subdivision is also part of the recently revisited concept of an ancestral division of the insect protocerebrum into archicerebrum (*six3*-positive) and prosocerebrum (*otd*-positive) [11]. As discussed above, *Tc-six3* is expressed in the anterior-most four type-II NB lineages, but unexpectedly, we found that *Tc-otd* is specifically absent from all type-II NB lineages, suggesting that its inhibition is required for the development of all these lineages. Interestingly, the lineages of type-II NB 5 and 6 (and presumably of type-II NB 7) of the anterior cluster and the posterior type-II NB lineages do neither express *Tc-otd* nor *Tc-six3* showing that there are protocerebral structures expressing none of these conserved markers in development, questioning the assigned generalized role of these two factors in subdividing the protocerebrum [11,55].

In summary, our findings are just the beginning of the quest for the genes required for the identity specification of type-II NB lineages. The genes known to be involved in head patterning are an excellent starting point for this purpose [11,18].

Methods

Creation of transgenic reporter line

Using CRISPR-Cas9 we have inserted a transgene into the *Tc-fez/erm* locus that includes *egfp* behind the *Tribolium* basal heat shock promoter (bhsp68), which is not active by itself but can be activated by nearby enhancer elements [64,65]. The transgene also contained the coding sequence of Cre-recombinase (not used in this work) transcribed from the same promoter, and the eye pigmentation gene *Tc-vermilion* behind the eye-specific 3xP3 enhancer element [66] (see Fig. S1 A for map of the transgene). On the repair plasmid the transgene was flanked by CRISPR target sequences derived from *Dm-yellow* and *Dm-ebony*, which were used for *in vivo* linearisation of the transgene.

Using CRISPR Optimal Target Finder [67] we designed three guide RNAs (gRNAs) targeting the genomic region 2.5 kb upstream of the *Tc-fez/erm* transcription start site (see table S1). We produced three plasmids where single gRNAs were transcribed from the *Tribolium* pU6b RNA promoter, as described in [12]. We co-injected the three gRNAs plasmids at 125 µg/µl each, gRNAs targeting the *Dm-yellow* and *Dm-ebony* sequence at 125 µg/µl each, Cas9 helper plasmid (see [12]) at 500 µg/µl and the repair plasmid at 500 µg/µl. Embryos of the white eyed *Tribolium* *vermilion-white* (*vw*) strain (G0) were injected, raised and then bred to *vw*-beetles in individual crosses. The offspring of these crosses (F1) were screened for black eyed heterozygous carriers of the transgene. These were again outcrossed to *vw*-beetles and the heterozygous offspring (F2) were further analysed genetically (see table S2) and with respect to fluorescent signal. Siblings of the F2 generation were then crossed to one another to generate the homozygous line (*fez-mm-eGFP*). Details of the insertion can be found in Fig. S1 A.

We also used the shaking hands (*skh*) enhancer trap line (G10011-GFP) which marks central complex cells [41].

Larval brain dissection, fixation, and staining

Larval brains of the newly generated *Tc-fez/erm* enhancer trap line *fez-mm-eGFP* were dissected and stained as described in [16]. Brains were mounted in *Vectashield* for microscopic inspection. See table S4 for a list of antibodies and staining reagents used.

Embryonic RNA *in situ* hybridisation, hybridisation chain reaction and antibody staining

Gene identifiers and primer sequences that were used for amplification can be found in table S3. The fragments were inserted into a pJet1.2 cloning vector. Standard RNA *in situ* probes were synthesised using the 5X Megascript T7 kit (Ambion) according to the manufacturers protocol. Embryo fixation and single colour RNA *in situ* chain reaction in combination with antibody staining to eGFP or phospho-histone H3 staining to mark mitoses was conducted as described in [68]. Embryos were also stained for DNA (dapi) or cell membranes (FM1-43). A full list of antibodies and staining reagents can be found in table S4. Probes for multicolour *hybridisation chain reaction* (HCR) binding *Tc-fez/erm*, *Tc-dpn*, *Tc-ase* and *Tc-pnt* were produced by Molecular Instruments (see table S5). Labelling reactions were performed as described in [69]. Antibody staining to eGFP was performed following the completion of the HCR staining (see [68]). All embryos were mounted in *Vectashield* for microscopic inspection (see table S4).

Image acquisition and analysis

Multichannel image stacks were recorded using a Zeiss LSM 980 confocal laser scanning microscope. Image stacks consisted of 100-300 slices, depending on the specimen. The resolution ranged from 1024 x 1024 to 2048 x 2048 pixels. Plane thickness was optimized according to width of the pinhole and ranged from 0.1 – 1 µm. We used FIJI [70] for 3 dimensional inspection, to export individual planes or to generate maximum intensity projections. Cropping and adjustment of brightness and contrast was done with GIMP (version 2.10.32) or Adobe Photoshop CS5.

Evaluation and of cell size and cell number

Cell and nuclear diameters of type-II NBs and a control group were measured based on the image stacks using FIJI [70] across 7 embryos. Lineage size and numbers of mitoses (based on *Tc-pnt* and *fez-mm-eGFP* expression/ anti-PH3 staining) was evaluated by manual counting based on image stacks derived from 4 head lobes of 4 embryos (= 28 anterior-medial lineages). The number of *Tc-dpn* expressing intermediate progenitors was evaluated in the same way across 5 embryos (= 35 anterior-medial lineages). Reference data on *Drosophila* lineage size (type-II NBs, INPs, GMCs and neurons) and numbers of *dpn*-expressing intermediate INPs were extracted from [29]. Statistics on all numerical data (standard deviations and two-tailed t-tests) were performed using Microsoft Excel 2016.

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Additional files

Supplementary material [↗](#)

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Reviewer #1 (Public review):

Summary:

Insects inhabit diverse environments and have neuroanatomical structures appropriate to each habitat. Although the molecular mechanism of insect neural development has been mainly studied in *Drosophila*, the beetle, *Tribolium castaneum* has been introduced as another model to understand the differences and similarities in the process of insect neural development. In this manuscript, the authors focused on the origin of the central complex. In

Drosophila, type II neuroblasts have been known as the origin of the central complex. Then, the authors tried to identify those cells in the beetle brain. They established a *Tribolium* *fez* enhancer trap line to visualize putative type II neuroblasts and successfully identified 9 of those cells. In addition, they also examined expression patterns of several genes that are known to be expressed in the type II neuroblasts or their lineage in *Drosophila*. They concluded that the putative type II neuroblasts they identified were type II neuroblasts because those cells showed characteristics of type II neuroblasts in terms of genetic codes, cell diameter, and cell lineage.

Strengths:

The authors established a useful enhancer trap line to visualize type II neuroblasts in *Tribolium* embryos. Using this tool, they have identified that there are 9 type II neuroblasts in the brain hemisphere during embryonic development. Since the enhancer trap line also visualized the lineage of those cells, the authors found that the lineage size of the type II neuroblasts in the beetle is larger than that in the fly. They also showed that several genetic markers are also expressed in the type II neuroblasts and their lineages as observed in *Drosophila*.

Comments on revisions:

The revisions have improved the manuscript greatly. However, I still have some concerns about the lack of examination of the expression of NB markers. Without examining the expression of at least one unequivocal neuroblast marker, no one can say confidently that it is a neuroblast. However, it is acknowledged that such a marker is currently not available for *Tribolium*.

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Reviewer #2 (Public review):

The authors address the question of differences in the development of the central complex (Cx), a brain structure mainly controlling spatial orientation and locomotion in insects, which can be traced back to the neuroblast lineages that produce the Cx structure. The lineages are called type-II neuroblast (NB) lineages and assumed to be conserved in insects. While *Tribolium castaneum* produces a functional larval Cx that only consists of one part of the adult Cx structure, the fan-shaped body, in *Drosophila melanogaster* a non-functional neuropile primordium is formed by neurons produced by the embryonic type-II NBs which then enter a dormant state and continue development in late larval and pupal stages.

The authors present a meticulous study demonstrating that type-II neuroblast (NB) lineages are indeed present in the developing brain of *Tribolium castaneum*. In contrast to type-I NB lineages, type-II NBs produce additional intermediate progenitors. The authors generate a fluorescent enhancer trap line called *fez/earmuff* which prominently labels the mushroom bodies but also the intermediate progenitors (INPs) of the type-II NB lineages. This is convincingly demonstrated by high resolution images that show cellular staining next to large pointed labelled cells, a marker for type-II NBs in *Drosophila melanogaster*. Using these and other markers (e.g. *deadpan*, *asense*), the authors show that the cell type composition and embryonic development of the type-II NB lineages are similar to their counterparts in *Drosophila melanogaster*. Furthermore, the expression of the *Drosophila* type-II NB lineage markers *six3* and *six4* in subsets of the *Tribolium* type-II NB lineages (anterior 1-4 and 1-6 type-II NB lineages) and the expression of the Cx marker *skh* in the distal part of most of the lineages provide further evidence that the identified NB lineages are equivalent to the *Drosophila* lineages that establish the central complex. However, in contrast to *Drosophila*, there are 9 instead of 8 embryonic type-II NB lineages per brain hemisphere and the lineages

contain more progenitor cells compared to the *Drosophila* lineages. The authors argue that the higher number of dividing progenitor cells supports the earlier development of a functional Cx in *Tribolium*.

While the manuscript clearly shows that type-II NB lineages similar to *Drosophila* exist in *Tribolium*, it does not establish a direct link between the characteristics of these lineages and a functional larval Cx in *Tribolium*, i.e., it does not identify the cause of the heterochronic development of the Cx in these insects. However, the detailed study lays the foundation for lineage tracing and gene function experiments that will elucidate if the higher number of *Tribolium* type-II NB lineage progenitors, the additional lineage and the timing of developmental progression of the progenitors can indeed be linked with the earlier function of the Cx and/or if other components are required for establishing the functional larval neural circuits in *Tribolium* such as e.g. larval born neurons as is the case in *Drosophila*.

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Reviewer #3 (Public review):

Summary:

In this paper, Rethemeier et al capitalize on their previous observation that the beetle central complex develops heterochronically compared to the fly and try to identify the developmental origin of this difference. For this reason, they use a *fez* enhancer trap line that they generated to study the neuronal stem cells (INPs) that give rise to the central complex. Using this line and staining against *Drosophila* type-II neuroblast markers, they elegantly dissect the number of developmental progression of the beetle type II neuroblasts. They show that the NBs, INPs, and GMCs have a conserved marker progression by comparing to *Drosophila* marker genes, although the expression of some of the lineage markers (*otd*, *six3*, and *six4*) is slightly different. Finally, they show that the beetle type II neuroblasts lineages are likely longer than the equivalent ones in *Drosophila* and argue that this might be the underlying reason for the observed heterochrony.

Strengths:

- Very interesting study system that compares a conserved structure that, however, develops in a heterochronic manner.
- Identification of a conserved molecular signature of type-II neuroblasts between beetles and flies. At the same time, identification of transcription factors expression differences in the neuroblasts, as well as identification of an extra neuroblast.
- Nice detailed experiments to describe the expression of conserved and divergent marker genes, including some lineaging looking into co-expression of progenitor (*fez*) and neuronal (*skh*) markers.

Weaknesses:

- The link between size and number of neuroblast lineages and the earlier central complex development in beetles is not examined.

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Author response:

The following is the authors' response to the original reviews.

General Response to Public Reviews

We thank the three reviewers for their positive evaluation of our work, which presents the first molecular characterization of type-II NB lineages in an insect outside the fly *Drosophila*. They seem convinced of our finding of an additional type-II NB and increased proliferation during embryogenesis in the red flour beetle. The reviewers expressed hesitations on our interpretation that the observed quantitative differences of embryonic lineages can directly be linked to the embryonic development of the central complex in *Tribolium*. While we still believe that a connection of both observations is a valid and likely hypothesis, we acknowledge that due the lack of functional experiments and lineage tracing a causal link has not directly been shown. We have therefore changed the manuscript to an even more careful wording that on one hand describes the correlation between increased embryonic proliferation with the earlier development of the Cx but on the other hand also stresses the need for additional functional and lineage tracing experiments to test this hypothesis. We have also strengthened the discussion on alternative explanations of the increased lineage size and emphasize the less disputed elements like presence and conservation of type-II NB lineages.

While our manuscript could in conclusion not directly show that the reason of the heterochronic shift lies in the progenitor behaviour, we still provide a first approach to answering the question of the developmental basis of this shift and testable hypotheses directly emerge from our work. We agree with reviewer#1 that functional work is best suited to test our hypothesis and we are planning to do so. However, we believe that the presented work is already rich in novel data and significantly advances our understanding on the conservation and divergence of type-II NBs in insects. We would also like to stress that most transgenic tools for which genome-wide collections exist for *Drosophila* have to be created for *Tribolium* and doing so can be quite time consuming. Conducting RNAi experiments is certainly possible in *Tribolium* but observing phenotypes in this defined cellular context will need laborious optimization. We have for example tried knocking down Tc-fez/erm but could not see any embryonic phenotype which might be due to an escaper effect in which only mildly affected or wild type-like embryos survive while the others die in early embryogenesis. Due to pleiotropic functions of the involved genes a cell-specific knockdown might be necessary and we are working towards establishing a system to do that in the red flour beetle. For the stated reasons, we see our work as an important basis to inspire future functional studies that build up on the framework that we introduced.

In response to these common points, we have made the following changes to the manuscript

- The title has been changed from 'being associated' to 'correlate'
- The conclusions part of the abstract has been changed
- We deleted the statement '...thus providing the material for the early central complex formation...'
- Rephrased to saying that the two observations just correlate
- The part of the discussion 'Divergent timing of type-II NB activity and heterochronic development of the central complex' has been extensively rewritten and now discusses several alternative explanations that were suggested by the reviewers. It also stresses the need for further functional work and lineage tracing (line 859-862 (608-611)).

In addition, we have made numerous changes to the manuscript to account for more specific comments of the reviewers and to the recommendations for the authors.

Our responses to the individual comments can be found in the following.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

*Insects inhabit diverse environments and have neuroanatomical structures appropriate to each habitat. Although the molecular mechanism of insect neural development has been mainly studied in *Drosophila*, the beetle, *Tribolium castaneum* has been introduced as another model to understand the differences and similarities in the process of insect neural development. In this manuscript, the authors focused on the origin of the central complex. In *Drosophila*, type II neuroblasts have been known as the origin of the central complex. Then, the authors tried to identify those cells in the beetle brain. They established a *Tribolium* fez enhancer trap line to visualize putative type II neuroblasts and successfully identified 9 of those cells. In addition, they also examined expression patterns of several genes that are known to be expressed in the type II neuroblasts or their lineage in *Drosophila*. They concluded that the putative type II neuroblasts they identified were type II neuroblasts because those cells showed characteristics of type II neuroblasts in terms of genetic codes, cell diameter, and cell lineage.*

Strengths:

*The authors established a useful enhancer trap line to visualize type II neuroblasts in *Tribolium* embryos. Using this tool, they have identified that there are 9 type II neuroblasts in the brain hemisphere during embryonic development. Since the enhancer trap line also visualized the lineage of those cells, the authors found that the lineage size of the type II neuroblasts in the beetle is larger than that in the fly. They also showed that several genetic markers are also expressed in the type II neuroblasts and their lineages as observed in *Drosophila*.*

Weaknesses:

*I recommend the authors reconstruct the manuscript because several parts of the present version are not logical. For example, the author should first examine the expression of *dpn*, a well-known marker of neuroblast. Without examining the expression of at least one neuroblast marker, no one can say confidently that it is a neuroblast. The purpose of this study is to understand what makes neuroanatomical differences between insects which is appropriate to their habitats. To obtain clues to the question, I think, functional analyses are necessary as well as descriptive analyses.*

The expression of an exclusive type-II neuroblast marker would indeed have been the most convincing evidence. However, *asense* is absent from type-II NBs and *deadpan* is not specific enough as it is expressed in many other cells of the developing protocerebrum. The gene *pointed*, although also expressed elsewhere, emerged as the most specific marker. Therefore, we start with *pointed* and *fez/erm* to describe the first appearance and developmental progression of the cells and then add further evidence that these cells are indeed type-II neuroblasts. Further evidence is provided in the following chapters. We have discussed the need for functional work in the general response.

Reviewer #2 (Public Review):

*The authors address the question of differences in the development of the central complex (Cx), a brain structure mainly controlling spatial orientation and locomotion in insects, which can be traced back to the neuroblast lineages that produce the Cx structure. The lineages are called type-II neuroblast (NB) lineages and are assumed to be conserved in insects. While *Tribolium castaneum* produces a functional larval Cx that*

only consists of one part of the adult Cx structure, the fan-shaped body, in *Drosophila melanogaster* a non-functional neuropile primordium is formed by neurons produced by the embryonic type-II NBs which then enter a dormant state and continue development in late larval and pupal stages.

The authors present a meticulous study demonstrating that type-II neuroblast (NB) lineages are indeed present in the developing brain of *Tribolium castaneum*. In contrast to type-I NB lineages, type-II NBs produce additional intermediate progenitors. The authors generate a fluorescent enhancer trap line called *fez/earmuff* which prominently labels the mushroom bodies but also the intermediate progenitors (INPs) of the type-II NB lineages. This is convincingly demonstrated by high-resolution images that show cellular staining next to large pointed labelled cells, a marker for type-II NBs in *Drosophila melanogaster*. Using these and other markers (e.g. *deadpan*, *asense*), the authors show that the cell type composition and embryonic development of the type-II NB lineages are similar to their counterparts in *Drosophila melanogaster*. Furthermore, the expression of the *Drosophila* type-II NB lineage markers *six3* and *six4* in subsets of the *Tribolium* type-II NB lineages (anterior 1-4 and 1-6 type-II NB lineages) and the expression of the Cx marker *skh* in the distal part of most of the lineages provide further evidence that the identified NB lineages are equivalent to the *Drosophila* lineages that establish the central complex. However, in contrast to *Drosophila*, there are 9 instead of 8 embryonic type-II NB lineages per brain hemisphere and the lineages contain more progenitor cells compared to the *Drosophila* lineages. The authors argue that the higher number of dividing progenitor cells supports the earlier development of a functional Cx in *Tribolium*.

While the manuscript clearly shows that type-II NB lineages similar to *Drosophila* exist in *Tribolium*, it does not considerably advance our understanding of the heterochronic development of the Cx in these insects. First of all, the contribution of these lineages to a functional larval Cx is not clear. For example, how do the described type-II NB lineages relate to the DM1-4 lineages that produce the columnar neurons of the Cx? What is the evidence that the embryonically produced type-II NB lineage neurons contribute to a functional larval Cx? The formation of functional circuits could rely on larval neurons (like in *Drosophila*) which would make a comparison of embryonic lineages less informative with respect to understanding the underlying variations of the developmental processes. Furthermore, the higher number of progenitors (and consequently neurons) in *Tribolium* could simply reflect the demand for a higher number of cells required to build the fan-shaped body compared to *Drosophila*. In addition, the larger lineages in *Tribolium*, including the higher number of INPs could be due to a greater number of NBs within the individual clusters, rather than a higher rate of proliferation of individual neuroblasts, as suggested. What is the evidence that there is only one NB per cluster? The presented schemes (Fig. 7/12) and description of the marker gene expression and classification of progenitor cells are inconsistent but indicate that NBs and immature INPs cannot be consistently distinguished.

We thank this reviewer for pointing out the inconsistency in our classification of cells within the lineages as one central part of our manuscript. These were due to a confusion in the used terms (young vs. immature). We have corrected this mistake and have changed the naming of the INP subtypes to immature-I and immature-II. We are confident that based on the analysed markers, type-II NBs and immature INPs can actually be distinguished with confidence.

We agree that a functional link of increased proliferation to heterochronic CX development is not shown although we consider it to be likely. As stated in the general response we have changed the manuscript to saying that the two observations (higher number of progenitors and larger lineages/more INPs) correlate but that a causal link can only be hypothesized for

the time being. At the same time, we have strengthened the discussion on alternative explanations.

We would like to remain with our statement of an increased number of embryonic progeny of *Tribolium* type-II NBs. We counted the total number of progenitor cells emerging from the anterior median cluster and divided this by the number of type II NBs in that cluster. Hence, the shown increased number of cells represents an average per NB but is not influenced by the increased number of NBs. On the same line, we have never seen indication for the presence of additional NBs within any cluster while one type-II NB is what we regularly found. Hence, we are confident that we know the number of respective NBs. The fact that the fly data included also neurons and was counted at a later stage indicates that the observed differences are actually minimum estimates.

We have discussed that based on the position and comparison to the grasshopper we believe that *Tribolium* type-II NB 1-4 contribute to the x, y, z and w tracts. To confirm this, lineage tracing experiments would be necessary, for which tools remain to be developed.

We agree that the role of larvally born neurons and the fate of *Tribolium* neuroblasts through the transition from embryo to larva and pupa need to be further studied.

Available data suggests that the adult fan shaped body in *Tribolium* does not hugely differ in size from the *Drosophila* counterpart, although no data in terms of cell number is available. In the larva, however, no fan shaped body or protocerebral bridge can be distinguished in flies while in beetle larvae, these structures are clearly developed. Hence, we think that it is more likely that differences observed in the embryo reflect differences in the larval central complex. We discuss the need for further investigation of larval stages.

The main difference between Tribolium and Drosophila Cx development with regards to the larval functionality might be that Drosophila type-II NB lineage-derived neurons undergo quiescence at the end of embryogenesis so that the development of the Cx is halted, while a developmental arrest does not occur in Tribolium. However, this needs to be confirmed (as the authors rightly observe).

Indeed, there is evidence that cells contributing to the CX go into quiescence in flies – hence, this certainly is one of the mechanisms. However, based on our data we would suggest that in addition, the balance of embryonic versus larval proliferation of type-II lineages is different between the two insects: The increased embryonic proliferation and development leads to a functional larval CX in beetles while in flies, postembryonic proliferation may be increased in order to catch up.

Reviewer #3 (Public Review):

Summary:

In this paper, Rethemeier et al capitalize on their previous observation that the beetle central complex develops heterochronically compared to the fly and try to identify the developmental origin of this difference. For this reason, they use a fez enhancer trap line that they generated to study the neuronal stem cells (INPs) that give rise to the central complex. Using this line and staining against Drosophila type-II neuroblast markers, they elegantly dissect the number of developmental progression of the beetle type II neuroblasts. They show that the NBs, INPs, and GMCs have a conserved marker progression by comparing to Drosophila marker genes, although the expression of some of the lineage markers (otd, six3, and six4) is slightly different. Finally, they show that the beetle type II neuroblast lineages are likely longer than the equivalent ones in Drosophila and argue that this might be the underlying reason for the observed heterochrony.

Strengths:

- A very interesting study system that compares a conserved structure that, however, develops in a heterochronic manner.
- Identification of a conserved molecular signature of type-II neuroblasts between beetles and flies. At the same time, identification of transcription factors expression differences in the neuroblasts, as well as identification of an extra neuroblast.
- Nice detailed experiments to describe the expression of conserved and divergent marker genes, including some lineaging looking into the co-expression of progenitor (*fez*) and neuronal (*skh*) markers.

Weaknesses:

- Comparing between different species is difficult as one doesn't know what the equivalent developmental stages are. How do the authors know when to compare the sizes of the lineages between *Drosophila* and *Tribolium*? Moreover, the fact that the authors recover more INPs and GMCs could also mean that the progenitors divide more slowly and, therefore, there is an accumulation of progenitors who have not undergone their programmed number of divisions.

We understand the difficulty of comparing stages between species, but we feel that our analysis is on the save side. At stages comparable with respect to overall embryonic development (retracting or retracted germband), the fly numbers are clearly smaller. To account for potential heterochronic shifts in NB activity, we have selected the stages to compare based on the criteria given: In *Drosophila* the number of INPs goes down after stage 16, meaning that they reach a peak at the selected stages. In *Tribolium* the chosen stages also reflect the phase when lineage size is larger than in all previous stages. Therefore, we believe that the conclusion that *Tribolium* has larger lineages and more INPs is well founded. Lineage size in *Tribolium* might further increase just before hatching (stage 15) but we were for technical reasons not able to look at this. As lineage size goes down in the last stage of *Drosophila* embryogenesis the number of INPs goes down and type-II NB enter quiescence, we think it is highly unlikely that the ratio between *Tribolium* and *Drosophila* INPs reverses at this stage, but a study of the behaviour of type-II NB in *Tribolium* and whether there is a stage of quiescence is still needed.

- The main conclusion that the earlier central complex development in beetles is due to the enhanced activity of the neuroblasts is very handwavy and is not the only possible conclusion from their data.

As discussed in the general response we have made several changes to the manuscript to account for this criticism and discuss alternative explanations for the observations.

- The argument for conserved patterns of gene expression between *Tribolium* and *Drosophila* type-II NBs, INPs, and GMCs is a bit circular, as the authors use *Drosophila* markers to identify the *Tribolium* cells.

We tested the hypothesis that in *Tribolium* there are type-II NBs with a molecular signature similar to flies. Our results are in line with that hypothesis. If *pointed* had not clearly marked cells with NB-morphology or *fez/erm* had not marked dividing cells adjacent to these NBs, we would have concluded that no such cells/lineages exist in the *Tribolium* embryo, or that central complex producing lineages exist but express different markers. Therefore, we regard this a valid scientific approach and hence find this argument not problematic.

An appraisal of whether the authors achieved their aims, and whether the results support their conclusions: Based on the above, I believe that the authors, despite advancing significantly, fall short of identifying the reasons for the divergent timing of central complex development between beetle and fly.

We agree that based on the available data, we cannot firmly make that link and we have changed the text accordingly.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

In addition to these descriptive analyses, functional analyses can be included. RNAi is highly effective in this beetle.

We agree that functional analyses of some of the studied genes and possible effects of gene knockdowns on the studied cell lineages and on central complex development could be highly informative. However, when studying specific cell types or organs these experiments are less straight forward than it may seem as knockdowns often lead to pleiotropic effects, sterility or lethality. All the genes involved are expressed in additional cells and may have essential functions there. Given the systemic RNAi of *Tribolium*, it is challenging to unequivocally assign phenotypes to one of the cell groups. Overcoming these challenges is often possible but needs extensive optimization. Our study, though descriptive is already rich in data and is the first description of NB-II lineages in *Tribolium* central complex development. We see it as a basis for future studies on central complex development that will include functional experiments.

(1) Introduction

For these reasons the beetle...

*Could you explain the differences in the habitats between *Tribolium* and *Drosophila*? or What is the biggest difference between these two species at the ecological aspect?*

We have added a short characterisation of the main differences.

The insect central complex is an anterior...

The author should explain why they focus on the structure.

Added

It is however not known how these temporal...

If the authors want to get the answer to the question, they need to conduct functional analyses.

While we agree with the importance of functional work (see above) we believe that detailed descriptions under the inclusion of molecular markers as presented here is very informative by itself for understanding developmental processes and sets the foundation for the analysis of mutant/RNAi- phenotypes in future studies.

CX - Central complex?

We have opted to not use this abbreviation anymore for clarity.

"because intermediate cycling progenitors have also been..."

Is the sentence correct?

We have included 'INPs' in the sentence to make clear what the comparison refers to and added a comma

"However, molecular characterization of such lineage in another..."

The authors should explain why molecular characterization is necessary.

We have done so

(2) Results

*a) Figure 8. Could you delineate the *skh*/eGFP expression region?*

We have added brackets to figure 1 panel A to indicate the extent of *skh* and other gene expressions within the lineages.

b) This section should be reorganized for better logical flow.

There certainly are different ways to organize this part and we have considered different structures of the results part. We eventually subjectively concluded that the chosen one is the best fit for our data (also see comment below on *dpn*-expression).

c) For the tables. The authors should mention what statistical analysis they have conducted.

The tables themselves are just listing the raw numbers. They are the basis for the graph in figure 9. Statistical tests (t-test) are mentioned in the legend of that figure and now also in the Methods sections.

*"We also found that the large *Tc-pnt*..."*

The authors could examine the mitotic index using an anti-pH3 antibody.

We have used the anti-pH3 antibody to detect mitoses (figure 3C, table 1 and 3) but as data on mitoses based on this antibody is only a snapshot it would require a lot of image data to reliably determine an index in this specific cells. While mitotic activity over time possibly combined with live imaging might be very interesting in this system also with regards to the timing of development, for this basic study we are satisfied with the statement that the type-II NB are indeed dividing at these stages.

"Based on their position by the end of embryogenesis..."

How can the authors conclude that they are neuroblasts without examining the expression of NB markers?

Type-II NB do not express *asense* as the key marker for type I neuroblasts. To corroborate our argument that the cells are neuroblasts we have used several criteria:

- We have used the same markers that are used in *Drosophila* to label type-II NBs (*pnt*, *dpn*, *six4*). We are not aware of any other marker that would be more specific.

- We have shown that these cells are larger and have larger nuclei than neighbouring cells and they are dividing

- We have shown that these cells through their INP lineages give rise to central complex neuropile

We believe that these features taken together leave little doubt that the described cells are indeed neuroblasts.

"We found that the cells they had assigned as..."

How did the authors distinguish that they are really neuroblasts?

We see the difficulty that we first describe the position and development of these cells (e.g. fig 3) and then add further evidence (cell size, additional marker *dpn*) that these are neuroblasts (also see above). However, without previous knowledge on position (and on *pnt* expression as the most specific marker) the type-II NB could not have been distinguished from other NBs based on cell size or expression of other markers.

"Conserved patterns of gene expression..."

*This must be the first (especially *dpn*).*

Dpn is not specific to type-II NB because it is also expressed in type-1 NBs, mature INPs and possibly other neural cells. It is therefore impossible to identify type-II NBs based on this gene alone. We therefore first used the most specific marker, *pnt*, in addition to adjacent *fez* expression to identify candidates for type-II lineages. Then we mapped expression of further genes on these lineages to support the interpretation (and show homology to the *Drosophila* lineages). Although of course the structure of a paper does not necessarily have to reflect the sequence in which experiments were done we would find putting *dpn* expression first misleading as it would not be clear why exactly a certain part of the expression should belong to type-II NB. Also, our *pnt-fez* expression data shows the position of the NB-II in the context of the whole head lobe whereas the other gene expressions are higher magnifications focussing on details. We therefore believe that the structure we chose best fits our data and the other reviewers seemed to find it acceptable as well.

"As type-II NBs contribute to central..."

*Before the sentence, the author could explain differences in the central complex structure between *Tribolium* and *Drosophila* in terms of cell number and tissue size.*

We have added references on the comparisons of tissue sizes, but unfortunately there is no *Tribolium* data that can be directly compared to available *Drosophila* resources in terms of cell number.

"We conclude that the embryonic development of..."

How did the authors conclude? They must explain their logic.

*Actually, before this sentence, I only found the description of the comparison between *Tribolium* NBs and *Drosophila* once.*

We agree that this conclusion is not fully evident from the presented data. We have therefore changed this part to stating that there is a correlation with the earlier central complex development described in *Tribolium*. See also response to the general reviewer comments.

"Hence, we wondered..."

The authors need to do a functional assessment of the genes they mentioned.

We agree that the goals originally stated at the beginning of this paragraph can only be achieved with functional experiments. We have therefore rephrased this part.

(3) Discussion

"A beetle enhancer trap line..."

This part should be moved elsewhere (it does not seem to be a discussion)

In accordance with this comment and reviewer#2's similar comment we have removed this section. We have added a statement on the importance of testing the expression of an enhancer trap line to the results part and an added the use of CRISPR-Cas9 for line generation to the introduction.

"We have identified a total..."

The authors emphasized that they discovered 9 type II NBs. The authors should clarify how important this it

We have added some discussion on the importance of this finding.

Dpn is a neural marker - Is this correct?

According to Bier et al 1992 (now added as reference) dpn is a pan-neural marker. Reviewer#2 also recommended calling dpn a neural marker.

"Previous work described a heterochronic..." - reference?

Reference have been added

"By contrast, we show that Tribolium..."

What about the number of neurons in the central complex in Tribolium and Drosophila?

Does the lineage size of type II NBs reflect the number?

Unfortunately, we do not have numbers for that.

Reviewer #2 (Recommendations For The Authors):

I recommend using page and line numbers to make reviewing and revising less timeconsuming.

We apologize for this oversight. We include a line numbering system into our resubmission.

(1) Abstract

"These neural stem cells are believed to be conserved among insects, but their molecular characteristics and their role in brain development in other insect neurogenetics models, such as the beetle Tribolium castaneum have so far not been studied."

I recommend explaining the importance of studying Tribolium with regard to the evolution of brain centres rather than just stating that data are lacking.

We have now emphasized the importance of Tribolium as model for the evolution of brain centres.

"Intriguingly, we found 9 type-II neuroblast lineages in the Tribolium embryo while Drosophila produces only 8 per brain hemisphere."

It should be made clear that the 9 lineages also refer to brain hemispheres.

We have added this information

(2) Introduction

I would remove the first paragraph of the introduction; the use of Tribolium as model representative for insects is too general. The authors should focus on the specific question, i.e. the introduction should start with paragraph 2.

While we can relate to the preference for short and concise writing, we feel that giving some background on Tribolium might be important as we expect that many of our readers might be primarily Drosophila researchers. Keeping this paragraph also seems in line with a recommendation of reviewer#1 to add some additional information on Tribolium ecology.

"Several NBs of the anterior-most part of the neuroectoderm contribute to the CX and compared..."

The abbreviation has not been introduced.

For clarity we have now opted to not use this abbreviation but to always spell out central complex.

"Several NBs of the anterior-most part of the neuroectoderm contribute to the CX and compared to the ventral ganglia produced by the trunk segments, it is of distinctively greater complexity..."

Puzzling statement. Why would you compare a brain center with ventral ganglia? I recommend removing this.

We have changed this statement to just emphasizing the complexity of the brain structure.

"The dramatically increased number of neural cells that are produced by individual type-II lineages, and the fact that one lineage can produce different types of neurons..." In my opinion, this statement is too vague and unprofessional in style. Instead of "dramatically increased" use numbers.

We have removed 'dramatically increased' and now give a numeric example.

"The dramatically increased number of neural cells that are produced by individual type-II lineages, and the fact that one lineage can produce different types of neurons, leads to the generation of increased neural complexity within the anterior insect brain when compared to the ventral nerve cord..."

I assume that this statement relates to the comparison of type I and II nb lineages. However, type I NB lineages also produce different types of neurons due to GMC

temporal identity, and neuronal hemi-lineage identity.

We have rephrased and tried to make clear that the second part of the statement is not specific to type-II NB only. In line with the comment above we have also removed the reference to the ventral nerve cord.

"In addition, in Drosophila brain tumours have been induced from type-II NBs lineages [34], opening up the possibility of modelling tumorigenesis in an invertebrate brain, thus making these lineages one of the most intriguing stem cell models in invertebrates [35,36]."

This statement is misplaced here; it should be mentioned at the start (if at all).

We have moved this statement up.

"However, molecular characterisation of such lineages in another insect but the fly and a thorough comparison of type-II NBs lineages and their sub-cell-types between fly and beetle are still lacking"

The background information should include what is known about type-II NB lineages in Tribolium, including marker gene expression, e.g. Farnworth et al.

We refer to He et al 2019, Farnworth et al 2020 and Garcia-Perez 2021. All these publications speculate about a contribution of type-II NBs to Tribolium central complex development but do not show evidence of it. As we emphasize throughout the manuscript, the present work is the first description of type-II NB in Tribolium.

"The ETS-transcription factor pointed (pnt) marks type-II NBs [40,41], which do not express the type-I NB marker asense (ase) but the pro-neural gene deadpan (dnp)" Deadpan is considered a pan-neural gene. To avoid confusion, I would remove "proneural" throughout.

We have done so throughout the manuscript.

"We further found that, like the type-II NBs itself, the youngest Tc-pnt-positive but fezmm-eGFP-negative INPs neither express Tc-ase (Fig. 5D, pink arrowheads)." What is the evidence that these are the youngest pnt positive cells? Position? This needs to be explained.

We have clarified that 'youngest pnt-positive cells' refers to the position of these cells close to the type-II NB.

"Therefore these neural markers can be used for a classification of type II NBs (Tc-pnt+, Tc-ase-), young INPs (Tc-pnt+, Tc-fez/erm-, Tc-ase-), immature INPs (Tc-pnt+, Tcfez/erm-, Tc-ase+), mature INPs (Tc-dpn+, Tc-ase+, Tc-fez/erm+, Tc-pros+), and GMCs (Tc-ase+, Tcfez/erm+, Tc-pros+, Tc-dpn). This classification is summarized in Fig. 7 A-B."

This is not the best classification and not in line with the schemes in Figure 7 - the young INPs are also immature. What is the difference? It needs to be explained what "mature" means (dividing?).

Thank you for pointing this out. We have corrected the error in this part that confused the two original groups (young and immature). To take the immaturity of both types of INPs into account we have then also changed our naming of INP subtypes into immature-I and immature-II and throughout the manuscript). Figure 7 and figure 12 were also changed

accordingly. While our classification is primarily based on gene expression the available data indicates that both types of immature INPs are not dividing, whereas mature INPs are. We have added a statement on that to this part.

"In beetles a single-unit functional central complex develops during embryogenesis while in flies the structure is postembryonic."

This statement is vague - the authors need to explain what is meant by "single-unit". The phrase "The structure is postembryonic" also needs more explanation. The Drosophila CX neuroblasts lineages originate in the embryo and the neurons form a commissural tract that becomes incorporated into the fan-shaped body of the Cx.

We have explained single-unit central complex and have improved our summary of known differences in central complex development between fly and beetle.

"To assess the size of the embryonic type-II NBs lineages in beetles we counted the Tc-fez/erm positive (fez-mm-eGFP) cells (INPs and GMCs) associated with a Tc-pnt-expressing type-II NBs of the anterior medial group (type-II NBs lineages 1-7). It is not clear what is meant by "with a Tc-pnt-expressing type-II NBs". Is this a typo?"

We have removed this bit.

(3) Discussion

I would remove the first paragraph "A beetle enhancer trap lines reflects Tc-fez/earmuff expression". This is a repetition of the methods rather than a discussion.

This part has been removed also in line with reviewer#1's comment.

(4) Figures

Figure 2

To which developing structure do the strongly labelled areas in Figure 2D correspond?

We believe that these areas from the protocerebrum including central complex, mushroom bodies and optic lobe. We have added this to the text and to the figure legend.

Figure 7

What do A and B represent? Different stages?

A and B show the same lineage but map the expression of different additional markers for clarity. We have added an explanation of this.

The classification contradicts the description in the section "Conserved patterns of gene expression mark Tribolium type-II NBs, different stages of INPs and GMCs" (last sentence) where young INPs are first in the sequence and described as pnt+, erm-, ase- and immature INPs as pnt+ erm+ and ase+.

We have corrected this mistake and changed the names of the subtypes into immatureI and immature-II (see above).

"We conclude that the evolutionary ancient six3 territory gives rise to the neuropile of the z, y, x and w tracts."

Please clarify if six3 is also expressed in the corresponding grasshopper NB lineages or if your conclusion is based on the comparison of Drosophila and Tribolium and you assume that this is the ancestral condition.

Six3 expression has not been studied in grasshoppers. Owing to the highly conserved nature of an anterior median six3 domain in arthropods and bilaterian animals in general, we would expect it to be expressed anterior-medially in grasshoppers as well. In *Drosophila* the gene is expressed in the anterior-medial embryonic region where the type-II NBs are expected to develop, but to our knowledge it has not been specifically studied which type-II NB lineages are located within this domain. We have clarified in our text that we do not claim that the origin of anterior-medial type-II NB 1-4 and the X,Y, Z and W lineages from the six3 territory is highly conserved but only the territory itself. As far as we know our work is the first to analyse the relationship of type-II lineages and the conserved head patterning genes *six3* and *otd*. We have added some clarification of this into this part of the discussion.

(5) Methods

The methods section should include the methods for cell counting, as well as cell and nuclei size measurements including statistics (e.g. how many embryos, how many NB lineages). The comparison of the Tribolium NB lineage cell numbers to published Drosophila data should include a brief description of the method used in Drosophila (in addition to the method used here in Tribolium) so that the reader can understand how the data compare.

We have added a separate section on this to the Methods part which also includes the criteria used in *Drosophila*. We have also included some more information to the results part on the inclusion of neurons in the *Drosophila* counts that may only be partially included in our numbers. This does however not change the results in terms of larger numbers of progenitor cells in *Tribolium*.

(6) Typos and minor errors

Abstract

"However, little is known on the developmental processes that create this diversity"

Change to ... little is known about

Changed.

NBs lineages

Change to NB lineages throughout.

We have used text search to find and replace all position where this was used erroneously,

Results

"Schematic drawing of expression different markers in type-II NB lineages.."

Schematic drawing of expression of different markers

Corrected

Discussion

"However, the type-II NB 7, which is we assigned to the anterior medial group but which..."

.... which we assigned....

corrected

".....might be the one that does not have a homologue in the fly embryo The identification of more..." Full stop missing.

Added.

"Adult like x, y, and w tracts as well as protocerebral bridge are...."

Change to "The adult like x, y, and w tracts as well as the protocerebral bridge are...."

This part has been removed with the rewriting of this paragraph.

Reviewer #3 (Recommendations For The Authors):

(1) Suggestions for improved or additional experiments, data, or analyses:

a) The analysis of nuclear size is wrong. The authors compare the largest cell of a cluster of cells with a number of random cells from the same brain. It is obvious that the largest cell of a cluster will be larger than the average cell of the same brain. A better control would be to compare the largest cell of the pnt+ cluster with the largest cell of a random sample of cells, although this also comes with biases. Personally, I have no doubt that the authors are looking at neuroblasts, based on the markers they are using, so I would recommend completely eliminating Figure 4.

We agree that we produced a somewhat biased and expected result when we select the largest cell of a cluster for size comparison. However, we found it important to show based on a larger sample that these cells are also statistically larger than the average cell of a brain, which we think our assessment shows. We do not claim that type-II NBs are the largest cells of a brain, or that they are larger than type-I NBs, therefore in a random sample there might be cells that are equally big (see also distribution of the control sample shown in figure 4, and we have added a note on this to the text). We are happy to hear that this reviewer has no doubts we are looking at neural stem cells. However, reviewer#1 did express some hesitations and therefore we think it is important to keep the information on cell size as part of our argument that we are indeed looking at type-II NBs (gene expression, cell size, dividing, part of a neural lineage).

b) The comparison of NB, INP, and GMC numbers between *Drosophila* and *Tribolium* (section "The *Tribolium* embryonic lineages of type-II NBs are larger and contain more mature INPs than those of *Drosophila*") compares an experiment that the authors did with published data. I would suggest that the authors repeat the *Drosophila* stainings and compare themselves to avoid cases of batch effects, inconsistent counting, etc.

None of the authors is a *Drosophila* expert or has any experience at working with this model and reassessing the lineage size would require a number of combinatorial staining. Therefore, we feel that using the published data produced by experts and which also includes repeat experiments is for us the more reliable approach.

c) In Figure 10, there are some otd+ GFP+ cells laterally. What are these?

We believe that these cells contribute to the eye anlagen. We have added this information to the legend.

| (2) *Minor corrections to the text and figures:*

| a) *There are some typos in the text: e.g. "pattering" in the abstract.*

We have carefully checked the text for typos and hope that we have found everything.

| b) *The referencing of figures in the text is inconsistent (eg "Figure 5 panel A" vs "Figure 5D" on page 12).*

We have checked throughout the manuscript and made sure to always refer to a panel correctly.

| c) *In Figure 3C, the white staining (anti-PH3) is not indicated in the Figure.*

The label has been added in the figure.

| d) *Moreover, in Figure 3, green is not very visible in the images.*

We have improved the colour intensity where possible.

| e) *In the figures, it might be better to outline the cells with color-coded dashed circles instead of using arrows.*

We think that this would obscure some details of the stainings and create a rather artificial representation. We also feel that doing this consistently in all our images is an amount of work not justified by the degree of expected improvement to the figures

NOTE: We are submitting a revised version of the supplementary material which only contains two minor changes: a headline was added to Table S4 (Antibodies and staining reagents) and a typo was corrected in line one of table S5 (TC to Tc).

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